香港中文大學(深圳)
The Chinese University of Hong Kong, Shenzhen

ECO6102 / 2026–27 Term 1 (Fall)

Lecture 3a: Linear Regression

Enjie (Jack) Ma

Lecture 3 · Monday 21 September 2026

Key Dates and Exam Format

Finalized — as in the syllabus; problem sets are due by 23:59, no late work

- **Problem Set 1**: due Sunday 11 October 2026 (not Friday 9)
- **Problem Set 2**: due Friday 23 October 2026
- **Midterm Exam (30%)**: Monday 26 October 2026, 09:00–11:00
- **Problem Set 3**: due Friday 11 December 2026
- **Review**: Monday 14 December 2026 (Week 15)
- **Final Exam (40%)**: Monday 21 December 2026, 09:00–11:00
- **Exam format**: two hours each; closed book and in person; one double-sided A4 cheat sheet is allowed

Participation (10%)

Course participation policy — completion and practice

- Credit is based on completion, not correctness.
- One missed quiz is allowed without penalty or documentation.
- Submit your name, UID and one short open-ended answer. The feedback question is optional.
- Questions give you practice with the reasoning and question styles used in exams.
- Complete the quiz in the classroom using WeChat; submit before the lecture ends.

Roadmap

Wooldridge Chapter 4

- OLS recap
- 4.1 Overview of the Single-Equation Linear Model
- 4.2 Asymptotic Properties of Ordinary Least Squares
  - 4.2.1 Consistency
  - 4.2.2 Asymptotic Inference Using Ordinary Least Squares
  - 4.2.3 Heteroskedasticity-Robust Inference

Roadmap

Wooldridge Recap — current location in blue

OLS recap

4.1 Overview of the Single-Equation Linear Model

4.2 Asymptotic Properties of Ordinary Least Squares

4.2.1 Consistency

4.2.2 Asymptotic Inference Using Ordinary Least Squares

4.2.3 Heteroskedasticity-Robust Inference

Ordinary least squares

Wooldridge 4.2.1, p. 57 — choose the line with the smallest sum of squared residuals

OBSERVED DATA

Student i: attendance x_i (percentage points), standardized final score y_i.

- A candidate line predicts $a + bx_i$; its residual is $y_i - a - bx_i$.

Ordinary least squares

Wooldridge 4.2.1, p. 57 — choose the line with the smallest sum of squared residuals

OBSERVED DATA

Student i: attendance x_i (percentage points), standardized final score y_i.

- A candidate line predicts $a + bx_i$; its residual is $y_i - a - bx_i$.
- **Ordinary least squares (OLS)** chooses the intercept and slope to minimize

$$SSR(a, b) = \sum_{i=1}^{N} (y_i - a - bx_i)^2.$$

▸ First-order conditions

The OLS solution

Introductory Wooldridge, 6e, 2.2; Wooldridge 4.2.1 — intercept and slope for one regressor

$$\hat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

- $\bar{x}$ and $\bar{y}$ are sample averages; the data must contain variation in x.

The OLS solution

Introductory Wooldridge, 6e, 2.2; Wooldridge 4.2.1 — intercept and slope for one regressor

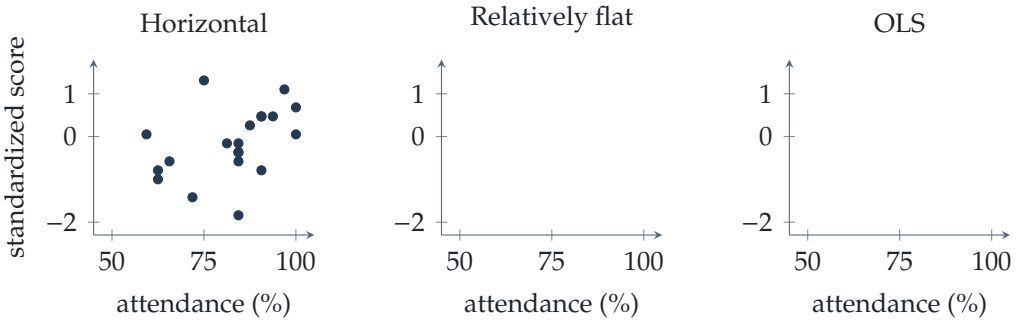
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- $\bar{x}$ and $\bar{y}$ are sample averages; the data must contain variation in x.
- The slope measures co-movement of x and y, scaled by variation in x.
- The intercept makes the fitted line pass through $(\bar{x}, \bar{y})$.

▸ Derivation ▸ Sample covariance and the limit

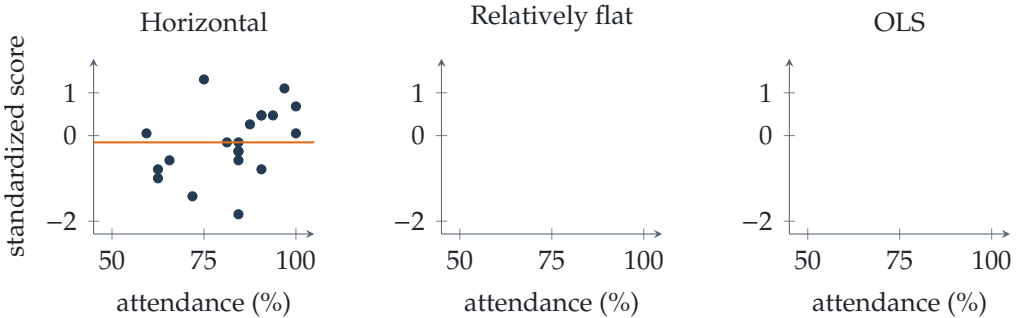
Candidate regression lines

Wooldridge 4.2.1 — the dots stay fixed; only the candidate line changes



Candidate regression lines

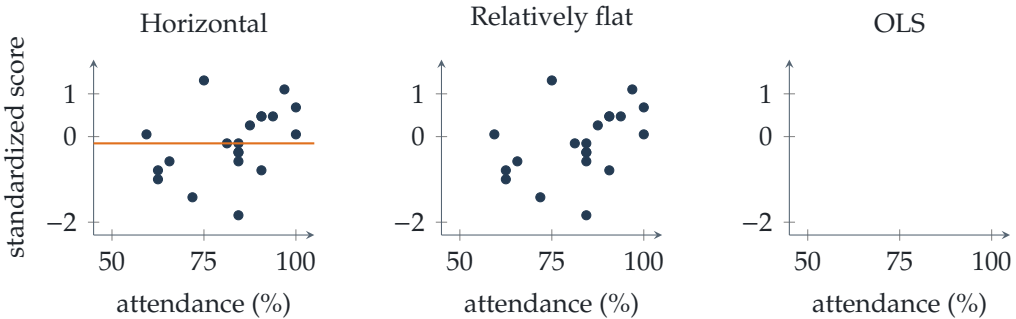
Wooldridge 4.2.1 — the dots stay fixed; only the candidate line changes



$a = -0.16,\ b = 0.0000$

Candidate regression lines

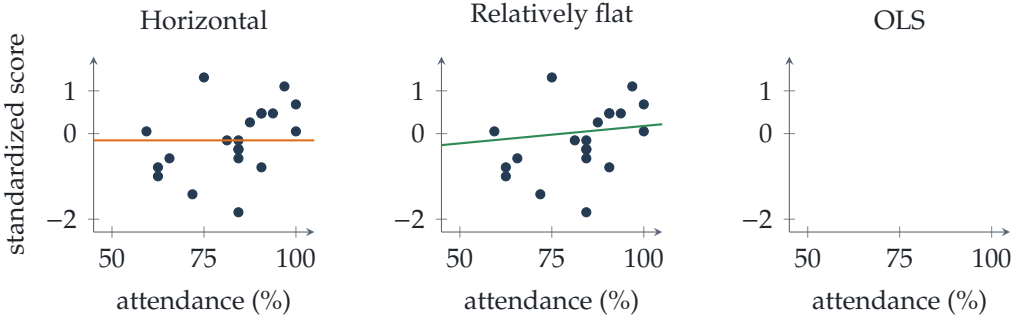
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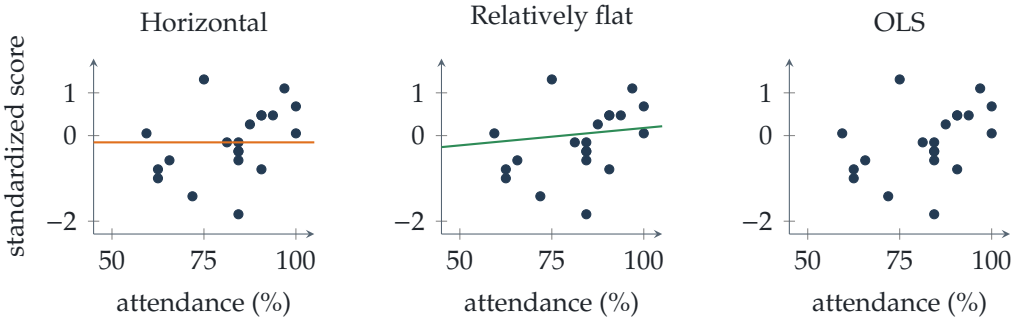
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$a = -0.16,\ b = 0.0000$ $a = -0.63,\ b = 0.0081$

Candidate regression lines

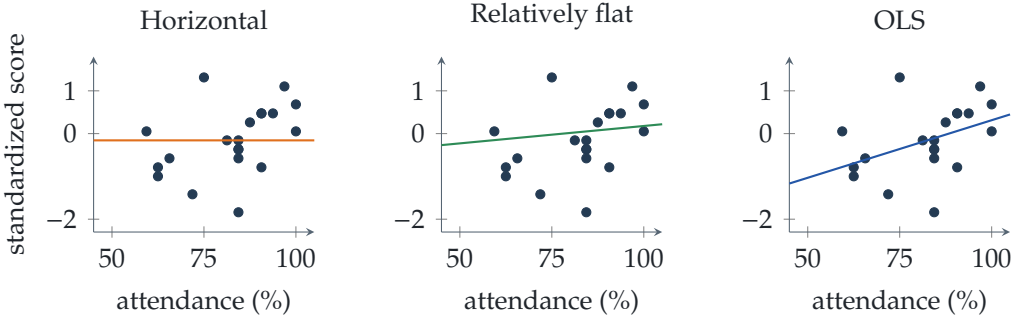
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Candidate regression lines

Wooldridge 4.2.1 — the dots stay fixed; only the candidate line changes



$a = -0.16,\ b = 0.0000$ $a = -0.63,\ b = 0.0081$ $a = -2.38,\ b = 0.0269$

Comparing squared residuals

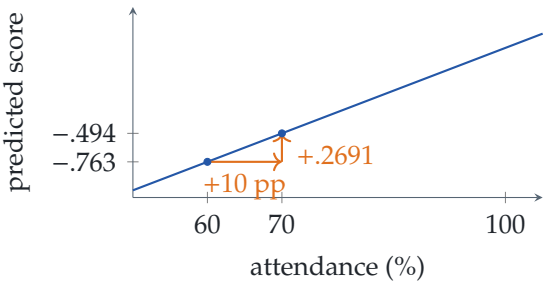
Wooldridge 4.2.1 — OLS wins this comparison by construction

Candidate	Line	SSR
Horizontal	$\hat{y} = -.1576$	12.27
Relatively flat	$\hat{y} = -.6345 + .00813x$	11.91
OLS	$\hat{y} = -2.3775 + .02691x$	10.11

- The last line minimizes squared residuals for the 20 observations just shown.

Interpreting the OLS slope

Wooldridge 4.2.1 — read the change in predicted score



- Same OLS line for the 20 students: slope .02691.
- Ten attendance percentage points go with about .2691 standard deviations of score.
- This is a fitted association. A causal interpretation needs more assumptions.

Population and sample regression functions

Wooldridge 4.1, (4.1)–(4.4) — a population relationship and a fitted sample line

- In a linear conditional-mean model, the **population regression function (PRF)** is

$$E(y \mid x) = \beta_0 + \beta_1 x, \qquad y = \beta_0 + \beta_1 x + u, \quad E(u \mid x) = 0.$$

Back to slide 20

Population and sample regression functions

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- In a linear conditional-mean model, the **population regression function (PRF)** is

$$E(y \mid x) = \beta_0 + \beta_1 x, \qquad y = \beta_0 + \beta_1 x + u, \quad E(u \mid x) = 0.$$

- The **sample regression function (SRF)** is the fitted line

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i, \qquad \hat{u}_i = y_i - \hat{y}_i.$$

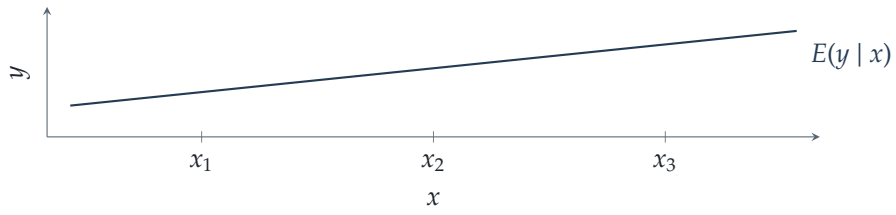
- The **error** u_i is measured from the population line; the **residual** $\hat{u}_i$ is measured from the fitted line.

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The population regression function

Wooldridge 2.2 and (4.4) — y is the conditional mean plus an individual deviation u

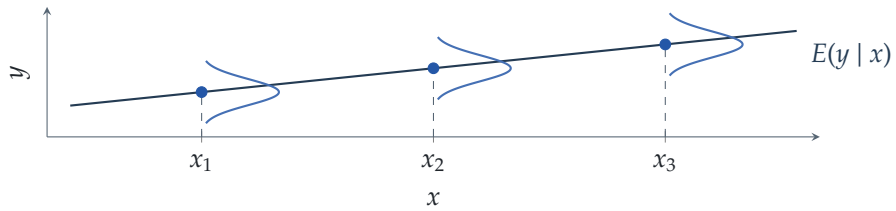
Solid line: **Population regression function (PRF)**.



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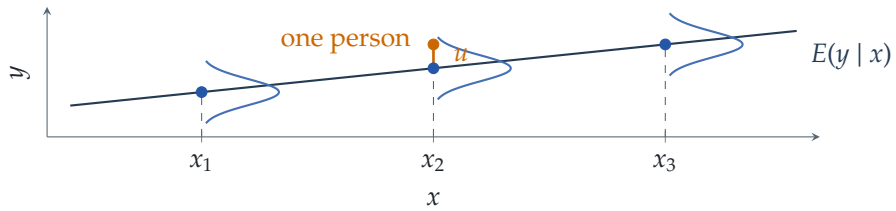


Each curve is the distribution of y among people with that x; its mean is the dot on the line, $E(y \mid x)$.

The population regression function

Wooldridge 2.2 and (4.4) — y is the conditional mean plus an individual deviation u

Solid line: **Population regression function (PRF)**.



Each curve is the distribution of y among people with that x; its mean is the dot on the line, $E(y \mid x)$. $y = E(y \mid x) + u$: the error is the gap to the *population* line; the residual, to the *fitted* line.

Parameter, estimator, and estimate

Wooldridge 3.5.1 and 4.2.1 — the target is fixed; the estimator changes with the sample

$$E(y \mid x) = \beta_0 + \beta_1 x, \qquad \hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x.$$

- **Parameter**: β_1, a fixed but unknown feature of the population model.
- **Estimator**: $\hat{\beta}_1$, a rule applied to random data; before observing the sample, its value is random.

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- **Parameter**: β_1, a fixed but unknown feature of the population model.
- **Estimator**: $\hat{\beta}_1$, a rule applied to random data; before observing the sample, its value is random.
- **Estimate**: the realized number from our data, such as .02691 for these 20 students.

Sampling variation and a sampling distribution

Wooldridge 3.5.1 — repeat the sampling rule and recompute the estimate

- **Sampling variation**: different random samples produce different estimates, even when the population and estimation rule stay the same.

‹ Back to slide 20

Sampling variation and a sampling distribution

Wooldridge 3.5.1 — repeat the sampling rule and recompute the estimate

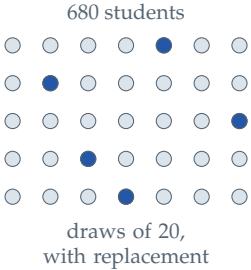
- **Sampling variation**: different random samples produce different estimates, even when the population and estimation rule stay the same.
- The **sampling distribution** describes those possible estimates and their probabilities at a given sample size.
- Our demonstration treats the 680 ATTEND students as an empirical population. Draw 20 students with replacement, fit OLS, and repeat.

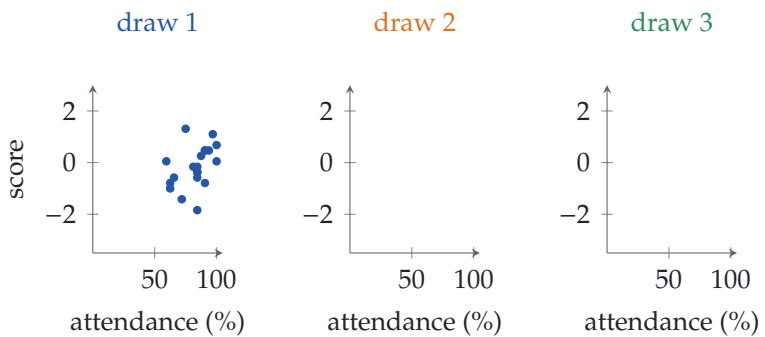
Each point in the next histogram is a slope from a whole sample, not one student's score.

‹ Back to slide 20

Three samples, three fitted lines

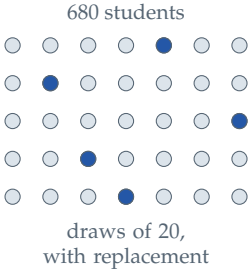
Wooldridge 4.2.1 — draw 20 students from the 680, refit, repeat: the data moved

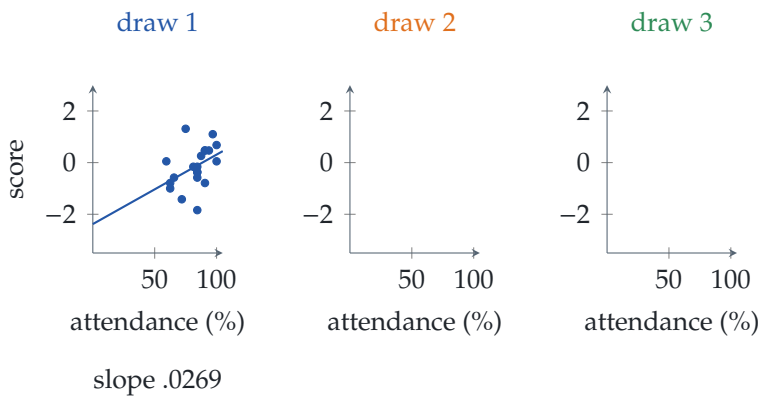




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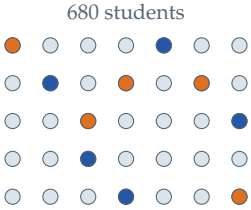
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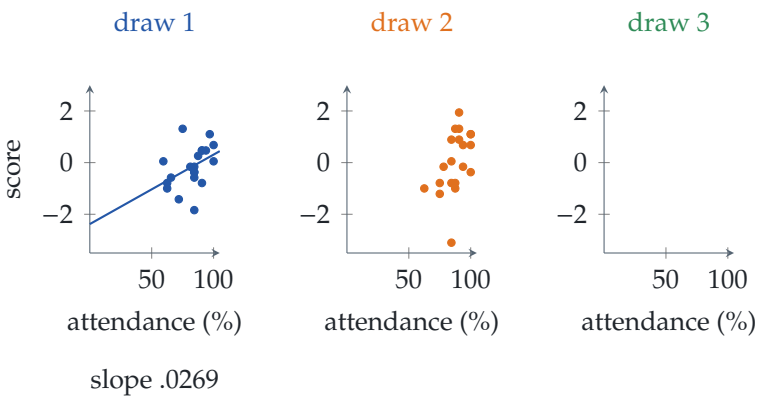




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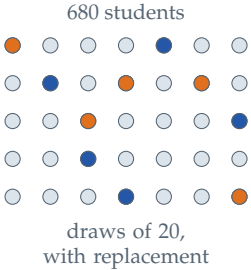
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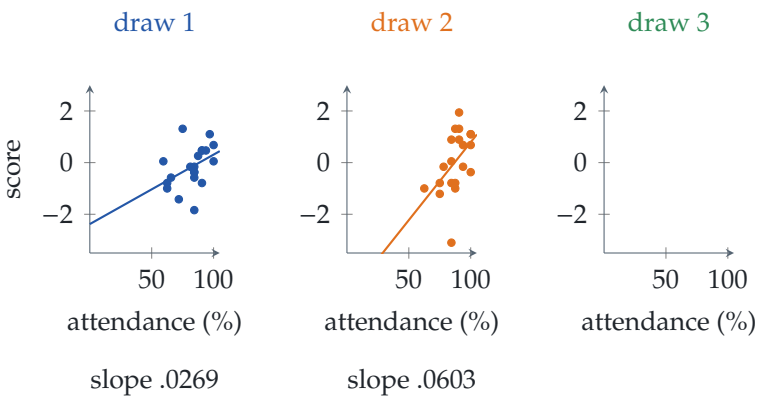




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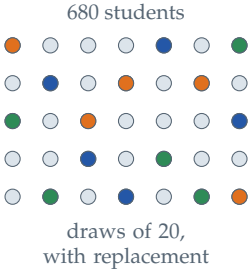
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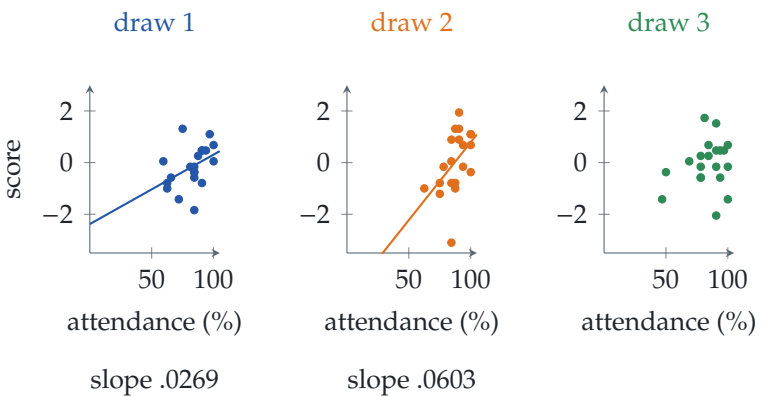




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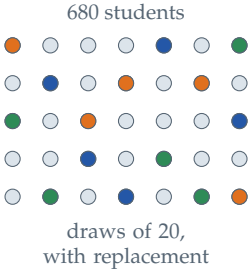
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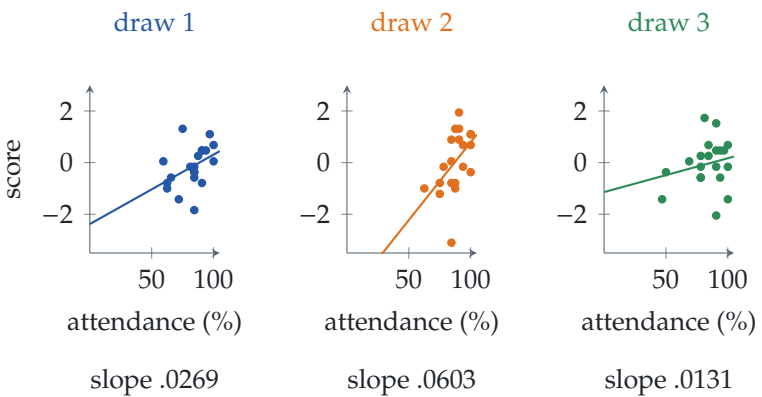




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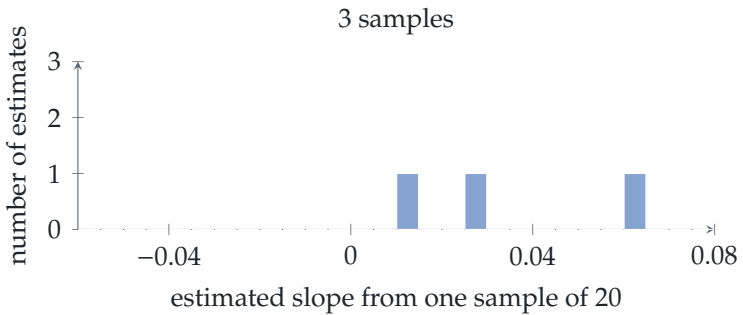
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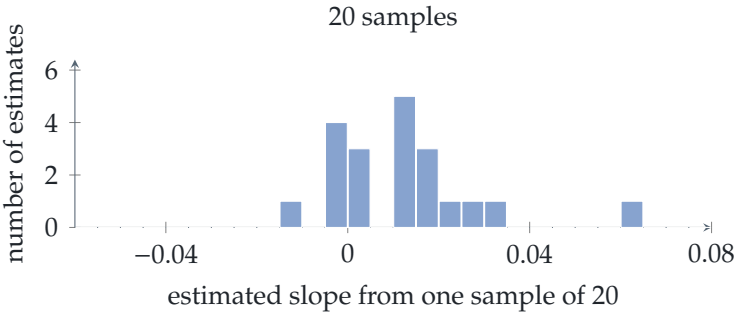
Build the sampling distribution

Wooldridge 3.5.1 — one sample contributes one estimated slope



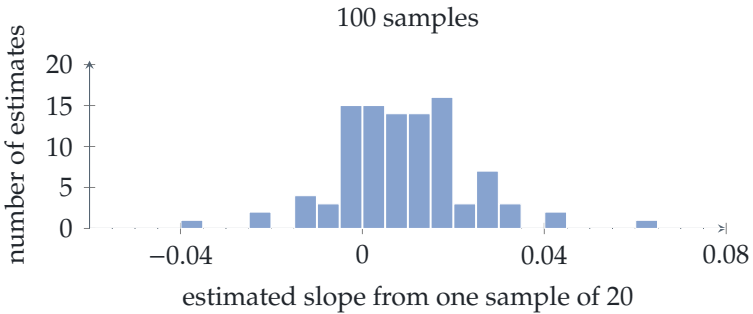
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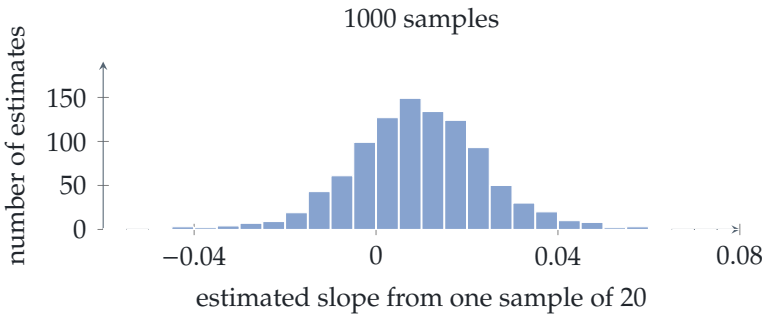
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Build the sampling distribution

Wooldridge 3.5.1 — one sample contributes one estimated slope



Centre: where estimates tend to land. Spread: how much they vary.

Bias compares an expectation with a target

Wooldridge 4.2.1, p. 58 — hold the sample size fixed

$$\text{Bias}(\hat{\beta}_N) = E(\hat{\beta}_N) - \beta; \qquad \text{unbiased: } E(\hat{\beta}_N) = \beta.$$

- Imagine every possible sample of the same size N.

Bias compares an expectation with a target

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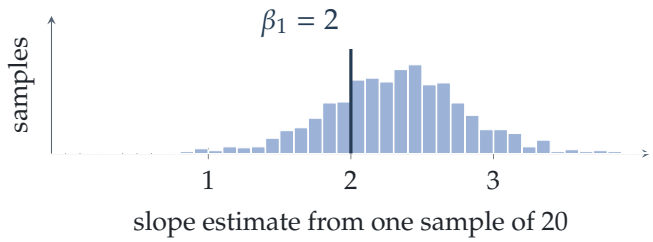
$$\text{Bias}(\hat{\beta}_N) = E(\hat{\beta}_N) - \beta; \qquad \text{unbiased: } E(\hat{\beta}_N) = \beta.$$

- Imagine every possible sample of the same size N.
- Average the estimates across those samples, then compare that expectation with the target.

Bias at a fixed sample size

Wooldridge 4.2.1 and Problem 4.2 — distinguish the centre from the spread

Illustrative: $y = 3 + 2x + u$, $E(u \mid x) = 0$, $N = 20$; $\tilde{\beta}_1 = \hat{\beta}_1 + 0.3$.

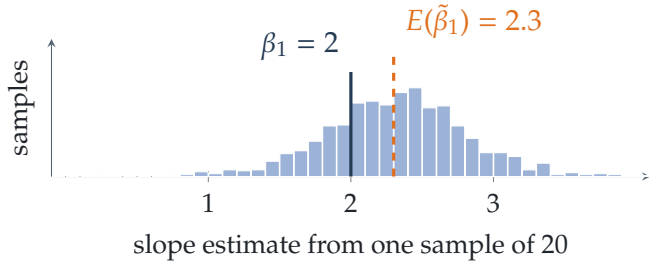


‣ Why OLS is unbiased here

Bias at a fixed sample size

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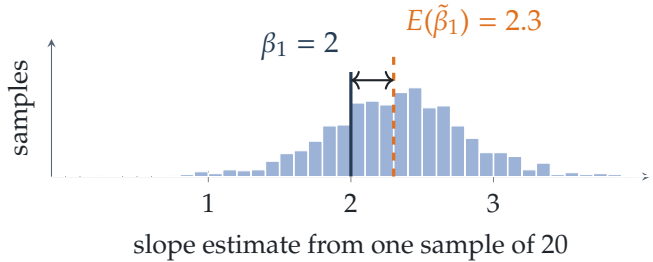


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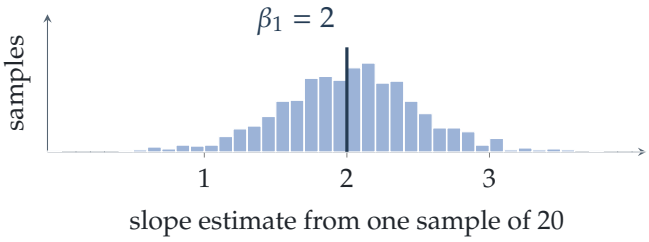


The expectation misses the target by 0.3: bias = .3. ▸ Why OLS is unbiased here

Unbiasedness at a fixed sample size

Wooldridge 4.2.1 and Problem 4.2 — distinguish the centre from the spread

Illustrative: $N = 20$, $y = 3 + 2x + u$, $E(u \mid x) = 0$.

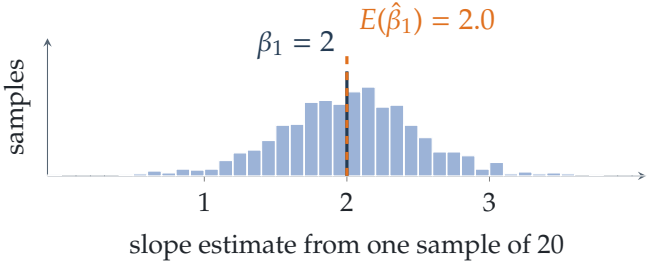


▸ Why OLS is unbiased here

Unbiasedness at a fixed sample size

Wooldridge 4.2.1 and Problem 4.2 — distinguish the centre from the spread

Illustrative: $N = 20$, $y = 3 + 2x + u$, $E(u \mid x) = 0$.



The two lines coincide: unbiased. Individual estimates still vary.

▸ Why OLS is unbiased here

What changes when the sample grows?

Wooldridge 3.5.1 — keep the population and estimation rule fixed

$$\hat{\beta}_N \xrightarrow{p} \beta \quad \Longleftrightarrow \quad P(|\hat{\beta}_N - \beta| > \varepsilon) \to 0 \quad \text{for every } \varepsilon > 0.$$

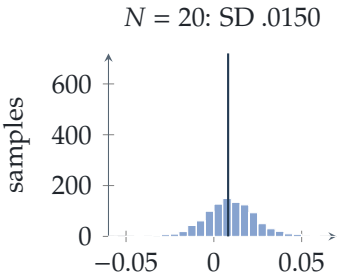
SAMPLING EXPERIMENT

We drew 20 students repeatedly. Now repeat the same experiment with 100 and then 500 students.

- Does collecting more data bring the estimate closer to our target?

Larger samples concentrate near the target

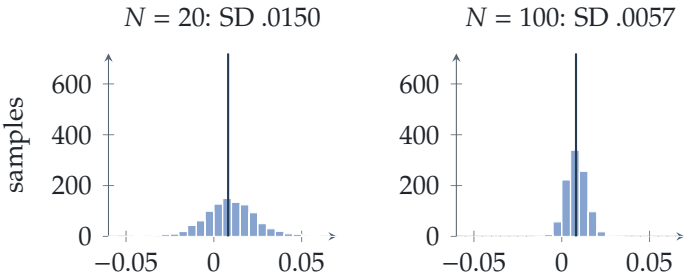
Wooldridge 3.5.1 — can I get closer to the truth just by collecting larger data?



Back to slide 23

Larger samples concentrate near the target

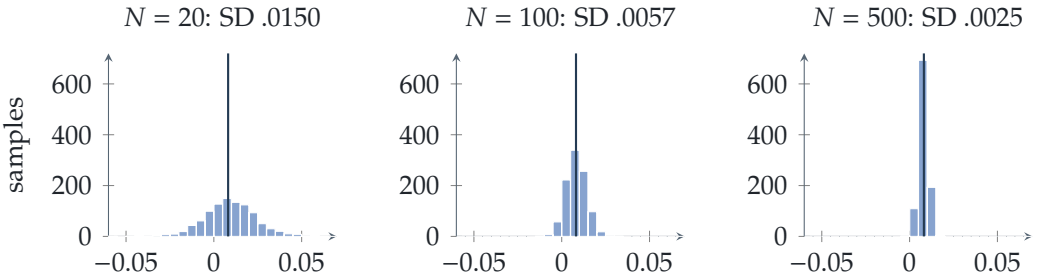
Wooldridge 3.5.1 — can I get closer to the truth just by collecting larger data?



Back to slide 23

Larger samples concentrate near the target

Wooldridge 3.5.1 — can I get closer to the truth just by collecting larger data?



At the same tolerance, more estimates lie near the target as N grows. The normal approximation will explain the scale of the spread.

Back to slide 23

OLS recap

Introductory Wooldridge, 6e, Chapters 2 and 5; Appendix C — three distinctions to carry forward

- **Fit and co-movement**: OLS chooses the smallest squared loss. A fitted slope measures association.
  - OLS solution: Chapter 2

OLS recap

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- **Fit and co-movement**: OLS chooses the smallest squared loss. A fitted slope measures association.
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- **Population and sample**: the target is fixed; another sample gives another estimate.
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- **Population and sample**: the target is fixed; another sample gives another estimate.
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- **Unbiasedness and consistency**: centered on the target at a fixed N versus concentrating near it as N grows.
  ‣ Compare the properties: Appendix C.2–C.3

Roadmap

Wooldridge 4.1 — current location in blue

OLS recap

4.1 Overview of the Single-Equation Linear Model

4.2 Asymptotic Properties of Ordinary Least Squares

4.2.1 Consistency

4.2.2 Asymptotic Inference Using Ordinary Least Squares

4.2.3 Heteroskedasticity-Robust Inference

The population equation and its error

Wooldridge 4.1, equation (4.1) — start with the relationship of interest

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p + u. \tag{4.1}$$

- The coefficients are population parameters; u contains unobserved determinants.

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$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p + u. \tag{4.1}$$

- The coefficients are population parameters; u contains unobserved determinants.
- **Linear in parameters**: regressors can include squares, logarithms and interactions.

A structural relationship

Wooldridge 4.1, pp. 53–54 — the coefficient has an economic meaning

QUESTION

How would an additional year of schooling change a person's wage, holding other relevant determinants fixed?

- A **structural relationship** represent a *causal* relationship.

A structural relationship

Wooldridge 4.1, pp. 53–54 — the coefficient has an economic meaning

QUESTION

How would an additional year of schooling change a person's wage, holding other relevant determinants fixed?

- A **structural relationship** represent a *causal* relationship.
- Here the target is the schooling effect at the same ability and experience, not simply the wage gap between schooling groups – *ceteris paribus*

From the target to credible estimation

Wooldridge 4.1–4.2.1 — target, comparison and consistency

SCHOOLING EQUATION

$$\log(wage) = \beta_0 + \beta_s\, schooling + \beta_e\, experience + u.$$

1. **Choose the target**: schooling's wage effect, holding relevant determinants fixed.

From the target to credible estimation

Wooldridge 4.1–4.2.1 — target, comparison and consistency

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$$E(u \mid schooling, experience) = 0.$$
3. **Ask for consistency**: with more observations, does our estimate concentrate near that target?

But how can we make sure what we estimate from data is consistent?

▸ Recall the consistency picture ▸ Formal definition of Consistency

Population restrictions for OLS

Wooldridge 4.1, equations (4.2)–(4.3) — the regressor must not move with the error

$$E(u) = 0, \qquad \mathrm{Cov}(x_j, u) = 0 \quad (j = 1, \ldots, p). \tag{4.2}$$

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- These restrictions concern the population, not fitted sample residuals.
- Zero conditional mean, $E(u \mid x_1, \ldots, x_p) = 0$, is a sufficient condition.
- With random sampling, finite moments and full population rank, (4.2) gives OLS consistency (Theorem 4.1).

▸ Conditional mean versus zero covariance ▸ Why the intercept handles the mean

Exogeneity vs. Endogeneity

Schooling example

- The error u contains wage determinants absent from the equation, such as unmeasured ability or motivation.
- A regressor is **exogenous** when it is uncorrelated with u; otherwise it is **endogenous**.

Exogeneity vs. Endogeneity

Schooling example

- The error u contains wage determinants absent from the equation, such as unmeasured ability or motivation.
- A regressor is **exogenous** when it is uncorrelated with u; otherwise it is **endogenous**.
- **Zero conditional mean** implies exogeneity: within every schooling-and-controls group, the omitted contribution to wages averages to zero.
- Then a higher-schooling group does not also have a higher mean omitted contribution to wages. This is an assumption, not a fact about schooling.

What could make schooling and omitted determinants move together?

▸ Zero conditional mean versus zero covariance

Three sources of endogeneity

Wooldridge 4.1, pp. 54–55 — how a regressor can become connected to the error

1. **Omitted variables**: a missing determinant affects both the regressor and the outcome.

Three sources of endogeneity

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Three sources of endogeneity

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1. **Omitted variables**: a missing determinant affects both the regressor and the outcome.
2. **Measurement error**: using an imperfect measure can put its measurement error into the regression error too.
3. **Simultaneity**: outcome conditions feed back into the regressor.

(1) Omitted variables

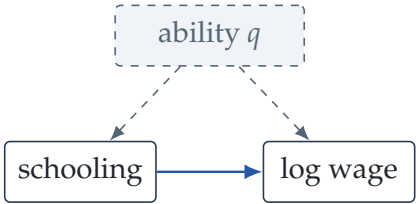
Wooldridge 4.1, pp. 54–55 — the omitted factor must affect both schooling and wages

- We want schooling's effect on wages, holding ability and experience fixed.
- Higher ability may raise wages and make further schooling more attractive.
- The observed wage gap can combine schooling's effect with an ability difference.

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- We want schooling's effect on wages, holding ability and experience fixed.
- Higher ability may raise wages and make further schooling more attractive.
- The observed wage gap can combine schooling's effect with an ability difference.
- This is *self-selection* into schooling



▸ Conditional means and ability

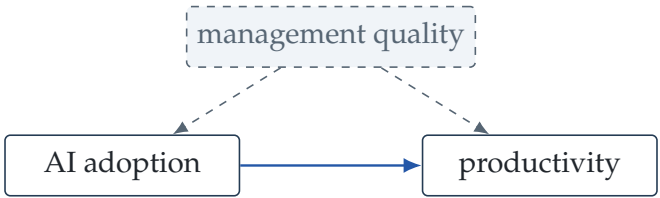
AI adoption, omitted variables, and self-selection

Wooldridge 4.1 — a hypothetical AI-adoption comparison

Firms using AI are more productive. Does this show that AI raised productivity, or can we not tell?

AI adoption, omitted variables, and self-selection

Wooldridge 4.1 — a hypothetical AI-adoption comparison



- Suppose better-managed firms are more likely to adopt and more productive for other reasons.
- If omitted management enters u, adoption and u may be correlated. A productivity gap alone does not isolate the tool's effect.

(2) Measurement error

Wooldridge 4.1, p. 55; 4.4.2 — a reporting mistake enters both parts of the comparison

A person reports more schooling than they actually completed.

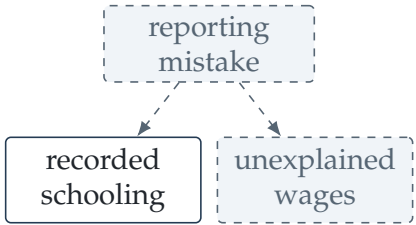
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- Their actual wages reflect true schooling. The unexplained part also contains the reporting mistake.

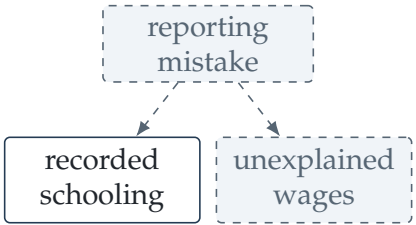


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CLASSICAL MEASUREMENT ERROR

Mean-zero reporting noise, uncorrelated with true schooling and the original wage disturbance.

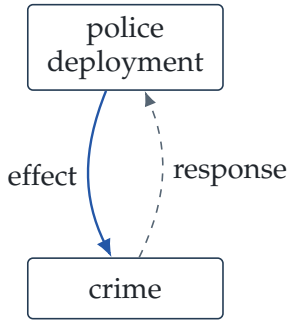
▸ Substitution and attenuation

(3) Simultaneity

Wooldridge 4.1, p. 55 — the regressor responds to determinants of the outcome

Simultaneity: the regressor and outcome are jointly determined.

- We want to know how adding police changes crime.



▸ Demand and supply example

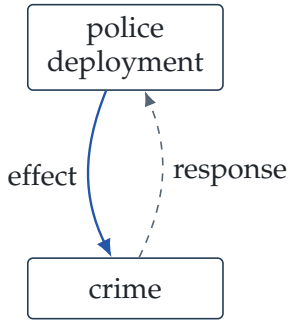
(3) Simultaneity

Wooldridge 4.1, p. 55 — the regressor responds to determinants of the outcome

Simultaneity: the regressor and outcome are jointly determined.

- We want to know how adding police changes crime.
- Police deployment responds to crime, which also depends on factors we have not observed.
- Police levels may therefore track unobserved crime determinants: police is endogenous.

▸ Demand and supply example



Three ways the comparison can go wrong

Wooldridge 4.1 — sources of endogeneity

- **Omitted variables**: schooling groups also differ in unobserved ability that matters for wages.

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Three ways the comparison can go wrong

Wooldridge 4.1 — sources of endogeneity

- **Omitted variables**: schooling groups also differ in unobserved ability that matters for wages.
- **Measurement error**: the recorded regressor and the regression error can share the same measurement mistake.
- **Simultaneity**: police deployment responds to crime conditions that we have not fully observed.

More observations alone cannot remove these connections. Proxies and instruments need additional assumptions.

One equation, three sources of endogeneity

Wooldridge 4.1, p. 55 — alcohol consumption and worker productivity

QUESTION

How does alcohol consumption affect productivity, measured by wages?

1. **Omitted variables**: family background may affect both alcohol use and wages.

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How does alcohol consumption affect productivity, measured by wages?

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QUESTION

How does alcohol consumption affect productivity, measured by wages?

1. **Omitted variables**: family background may affect both alcohol use and wages.
2. **Simultaneity**: income, largely determined by wages, can affect alcohol demand.
3. **Measurement error**: reported alcohol consumption may differ from actual use.

The distinctions are not always sharp; more than one source can occur in the same equation.

A sidenote: Structural vs. Reduced-form

Two perspectives of applied economics research

- **Design-based (often called reduced-form)**: use a credible comparison to estimate a policy or treatment effect, without estimating a full model of behavior.

A sidenote: Structural vs. Reduced-form

Two perspectives of applied economics research

- **Design-based (often called reduced-form)**: use a credible comparison to estimate a policy or treatment effect, without estimating a full model of behavior.
- **Structural estimation**: combine data with an economic model to estimate behavioral parameters and study counterfactuals.

Both need identifying assumptions; the approaches can inform each other.

▸ Policy-question example ▸ Formal reduced form ▸ Reading

Roadmap

Wooldridge 4.2 — current location in blue

OLS recap

4.1 Overview of the Single-Equation Linear Model

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- 4.2.1 Consistency
- 4.2.2 Asymptotic Inference Using Ordinary Least Squares
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The model in vector form

Wooldridge 4.2, (4.5) — a compact name for the same regression

$$y = \mathbf{x}\boldsymbol{\beta} + u. \tag{4.5}$$

- $\mathbf{x}$ collects the constant and regressors; $\boldsymbol{\beta}$ collects their coefficients.
- For attendance, $\mathbf{x}_i\boldsymbol{\beta} = \beta_0 + \beta_1 atndrte_i$.

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- For attendance, $\mathbf{x}_i\boldsymbol{\beta} = \beta_0 + \beta_1 atndrte_i$.
- N counts observations; K counts coefficients, including the intercept. The notation lets us handle several regressors together.

▸ Rows and columns with real data

Asymptotic properties

Wooldridge 4.2 introduction; Chapter 3 — let the sample grow under the same model

SAMPLING SETUP

A random sample $(\mathbf{x}_i, y_i)$, $i = 1, \ldots, N$, from one population.

- **Consistency**: does the estimate concentrate near the target as N grows?

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A random sample $(\mathbf{x}_i, y_i)$, $i = 1, \ldots, N$, from one population.

- **Consistency**: does the estimate concentrate near the target as N grows?
- **Asymptotic inference**: what distribution and spread approximate its sampling uncertainty?

▸ Mathematical definition of consistency ▸ LLN and CLT

Roadmap

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OLS recap

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Assumption OLS.1

Wooldridge 4.2.1, p. 56 — population errors must be unrelated to regressors

$$E(\mathbf{x}'u) = \mathbf{0}.$$

- The error has mean zero and is uncorrelated with every regressor in the population.

Assumption OLS.1

Wooldridge 4.2.1, p. 56 — population errors must be unrelated to regressors

$$E(\mathbf{x}'u) = \mathbf{0}.$$

- The error has mean zero and is uncorrelated with every regressor in the population.
- In the schooling example, omitted wage determinants must not systematically move with schooling or the included controls.

▸ Unpack OLS.1 ▸ Assumption versus fitted identity

Assumption OLS.2

Wooldridge 4.2.1, p. 57 — the data must separate the coefficients

$$\operatorname{rank} E(\mathbf{x}'\mathbf{x}) = K.$$

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Wooldridge 4.2.1, p. 57 — the data must separate the coefficients

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- No regressor can be an exact linear combination of the others in the population.
- Example: including age in both years and months cannot identify two separate effects; one column is twelve times the other.
- Correlated regressors are allowed. Perfect redundancy is the problem here.

▸ Rank condition and other examples

Identification comes before estimation

Wooldridge 4.2.1, p. 57 — express the coefficient in observable population moments

- In this linear model with random sampling, **identification** means that $\boldsymbol{\beta}$ can be written in terms of population moments of observable variables.

Identification comes before estimation

Wooldridge 4.2.1, p. 57 — express the coefficient in observable population moments

- In this linear model with random sampling, **identification** means that $\boldsymbol{\beta}$ can be written in terms of population moments of observable variables.
- OLS.1 and OLS.2 give

$$\boldsymbol{\beta} = [E(\mathbf{x}'\mathbf{x})]^{-1}E(\mathbf{x}'y).$$

- We observe $(\mathbf{x}, y)$; both expectations involve these observable variables.

Next, replace the population moments with their sample counterparts.

The analogy principle

Wooldridge 4.2.1 — replace population averages with sample averages

Population

$$\boldsymbol{\beta} = [E(\mathbf{x}'\mathbf{x})]^{-1} E(\mathbf{x}'y).$$

Sample

$$\hat{\boldsymbol{\beta}} = \left[\frac{1}{N}\sum_i \mathbf{x}_i'\mathbf{x}_i\right]^{-1}\left[\frac{1}{N}\sum_i \mathbf{x}_i' y_i\right].$$

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- We cannot see the whole population. We calculate averages and co-movement from our sample instead.
- The **analogy principle** solves the sample version of the population problem. Here it gives the same OLS fit as least squares.

▸ Population solution ▸ Sample solution ▸ Law of large numbers

Consistency of OLS

Wooldridge 4.2.1, Theorem 4.1 — the chance of a large miss tends to zero

$$\operatorname{plim}_{N\to\infty} \hat{\boldsymbol{\beta}} = \boldsymbol{\beta}.$$

- **Theorem 4.1**: under OLS.1–OLS.2, OLS from a random sample following $y = \mathbf{x}\boldsymbol{\beta} + u$ is consistent for $\boldsymbol{\beta}$.

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- **Theorem 4.1**: under OLS.1–OLS.2, OLS from a random sample following $y = \mathbf{x}\boldsymbol{\beta} + u$ is consistent for $\boldsymbol{\beta}$.
- As the sample grows, the probability of a large estimation error tends to zero.
- **Consistency** is about growing N. **Unbiasedness** is about being centered on the target at a fixed N; it is a different property.

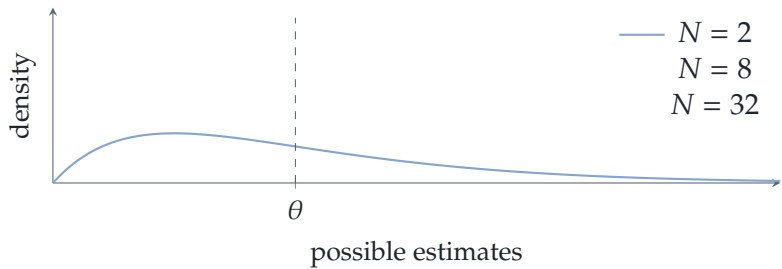
The next picture holds the target fixed and lets the sample size grow.

▸ Consistency proof ▸ Unbiasedness proof ▸ Definition of Consistency

▸ Unbiasedness and consistency

Consistency

Introductory Wooldridge 6e, 5-1, Figure 5.1 — schematic sampling distributions as the sample grows

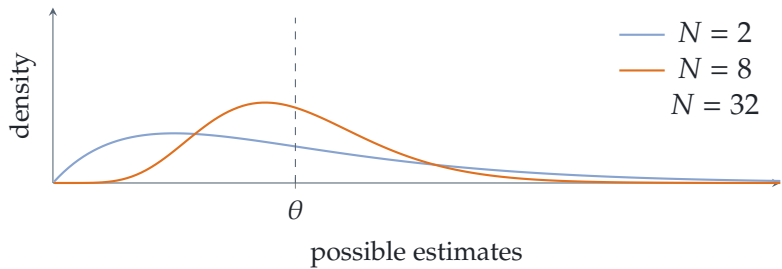


More data makes a large miss less likely; it need not improve every realized estimate.

▸ The probability definition

Consistency

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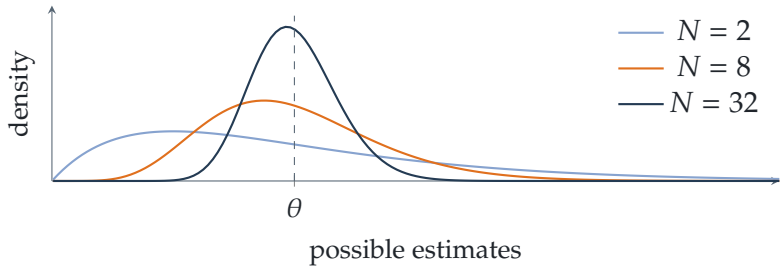


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Introductory Wooldridge 6e, 5-1, Figure 5.1 — schematic sampling distributions as the sample grows



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Roadmap

Wooldridge 4.2.2 — current location in blue

OLS recap

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Asymptotic normality

Wooldridge 4.2.2, p. 59 — the distribution needed for inference

- Consistency tells us that large misses become unlikely. Hypothesis testing also requires the asymptotic distribution of the estimator.

Asymptotic normality

Wooldridge 4.2.2, p. 59 — the distribution needed for inference

- Consistency tells us that large misses become unlikely. Hypothesis testing also requires the asymptotic distribution of the estimator.
- Under i.i.d. sampling, OLS.1–OLS.2 and finite second moments of $\mathbf{x}'u$,

$$\sqrt{N}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) \xrightarrow{d} N(\mathbf{0}, \mathbf{A}^{-1}\mathbf{B}\mathbf{A}^{-1}),$$

  where $\mathbf{A} = E(\mathbf{x}'\mathbf{x})$ and $\mathbf{B} = E(u^2\mathbf{x}'\mathbf{x})$.

▸ Derivation ▸ CLT intuition

Homoskedasticity and Theorem 4.2

Wooldridge 4.2.2, p. 59, (4.9) — simplify the asymptotic covariance

- **Assumption OLS.3**:

$$E(u^2\mathbf{x}'\mathbf{x}) = \sigma^2 E(\mathbf{x}'\mathbf{x}), \qquad \sigma^2 = E(u^2).$$

Homoskedasticity and Theorem 4.2

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- **Assumption OLS.3**:

$$E(u^2\mathbf{x}'\mathbf{x}) = \sigma^2 E(\mathbf{x}'\mathbf{x}), \qquad \sigma^2 = E(u^2).$$

- **Theorem 4.2**: under OLS.1–OLS.3 and the maintained sampling and moment conditions,

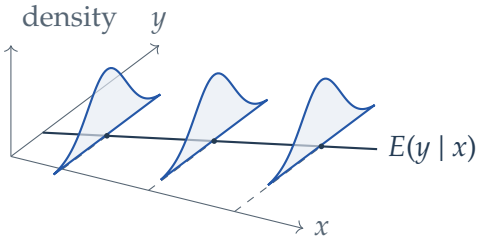
$$\sqrt{N}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) \xrightarrow{d} N\left(\mathbf{0}, \sigma^2 [E(\mathbf{x}'\mathbf{x})]^{-1}\right). \tag{4.9}$$

OLS.3 gives $\mathbf{B} = \sigma^2\mathbf{A}$, so $\mathbf{A}^{-1}\mathbf{B}\mathbf{A}^{-1} = \sigma^2\mathbf{A}^{-1}$.

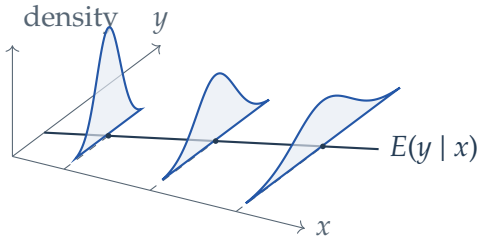
▸ Read OLS.3 element by element ▸ Proof

A conditional-variance interpretation

Wooldridge 4.2.2, p. 59 — a familiar sufficient condition for OLS.3



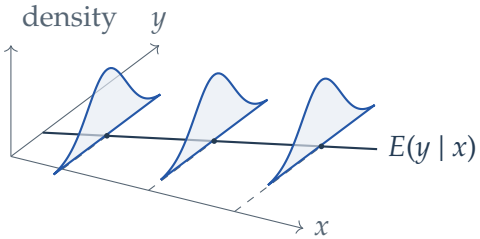
homoskedastic: constant spread



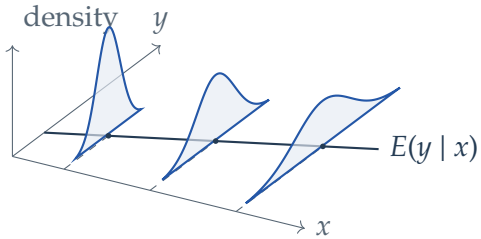
heteroskedastic: spread grows

A conditional-variance interpretation

Wooldridge 4.2.2, p. 59 — a familiar sufficient condition for OLS.3



homoskedastic: constant spread



heteroskedastic: spread grows

When $E(u \mid x) = 0$, $E(u^2 \mid x) = \mathrm{Var}(u \mid x)$.

Constant variance $\Longrightarrow$ OLS.3 (sufficient, not necessary).

▸ Why LIE gives OLS.3

Asymptotic variance under homoskedasticity

Wooldridge 4.2.2, p. 60 — from the limit to an estimated covariance

- Theorem 4.2 gives

$$\text{Avar}(\hat{\boldsymbol{\beta}}) = \frac{\sigma^2}{N}[E(\mathbf{x}'\mathbf{x})]^{-1}.$$

Asymptotic variance under homoskedasticity

Wooldridge 4.2.2, p. 60 — from the limit to an estimated covariance

- Theorem 4.2 gives

$$\text{Avar}(\hat{\boldsymbol{\beta}}) = \frac{\sigma^2}{N}[E(\mathbf{x}'\mathbf{x})]^{-1}.$$

- Estimate the unknown population quantities from the sample:

$$\hat{\sigma}^2 = \frac{\sum_i \hat{u}_i^2}{N-K}, \qquad \hat{\mathbf{A}} = \frac{\mathbf{X}'\mathbf{X}}{N}, \qquad \hat{u}_i = y_i - \mathbf{x}_i\hat{\boldsymbol{\beta}}.$$

K includes the intercept. The factor $1/N$ belongs to the covariance of $\hat{\boldsymbol{\beta}}$.

From estimated covariance to standard errors

Wooldridge 4.2.2, (4.10); Definition 3.10, p. 44 — one standard error per coefficient

SAMPLE INPUTS

$\hat{\sigma}^2 = SSR/(N-K)$ and $\hat{\mathbf{A}} = \mathbf{X}'\mathbf{X}/N$.

$$\widehat{\text{Avar}}(\hat{\boldsymbol{\beta}}) = \frac{\hat{\sigma}^2}{N}\hat{\mathbf{A}}^{-1} = \hat{\sigma}^2(\mathbf{X}'\mathbf{X})^{-1}. \tag{4.10}$$

From estimated covariance to standard errors

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$$\widehat{\text{Avar}}(\hat{\boldsymbol{\beta}}) = \frac{\hat{\sigma}^2}{N}\hat{\mathbf{A}}^{-1} = \hat{\sigma}^2(\mathbf{X}'\mathbf{X})^{-1}. \tag{4.10}$$

- The **standard error** is the square root of the corresponding diagonal element:

$$\text{se}(\hat{\beta}_j) = \sqrt{\left[\widehat{\text{Avar}}(\hat{\boldsymbol{\beta}})\right]_{jj}}.$$

- It estimates the sampling spread of that coefficient estimator.

▸ SE versus sampling SD ▸ Why the residual variance is consistent

Wooldridge 4.2.2

Statistical inference

Why the OLS results make inference possible

Wooldridge 3.5.1, pp. 43–44; 4.2.2, pp. 59–60 — connect assumptions, distribution and decisions

1. OLS.1–OLS.2 give consistency under random sampling.
2. Theorem 4.2 supplies a normal approximation; OLS.3 gives its usual covariance.

Why the OLS results make inference possible

Wooldridge 3.5.1, pp. 43–44; 4.2.2, pp. 59–60 — connect assumptions, distribution and decisions

1. OLS.1–OLS.2 give consistency under random sampling.
2. Theorem 4.2 supplies a normal approximation; OLS.3 gives its usual covariance.
3. A consistent covariance estimate supplies the scale:

$$\frac{\hat{\beta}_j - \beta_j}{\text{se}(\hat{\beta}_j)} \xrightarrow{d} N(0,1).$$

▸ Why standardizing gives this limit

Testing one coefficient

Wooldridge 4.2.2, p. 60 — compare a statistic with its null reference

$$H_0 : \beta_j = \beta_{j,0} \qquad \text{against} \qquad H_1 : \beta_j \neq \beta_{j,0}.$$

$$t = \frac{\hat{\beta}_j - \beta_{j,0}}{\text{se}(\hat{\beta}_j)} \overset{a}{\sim} N(0,1) \quad \text{under } H_0.$$

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- A two-sided 5% test rejects H_0 when $|t| > 1.96$.
- Otherwise, fail to reject H_0: insufficient evidence against the hypothesized value.

▸ Possible testing errors ▸ P-values

Statistical inference

Wooldridge 4.2.2; Example 4.1 data — read an estimate together with its uncertainty

MROZ: 428 working, married women.

$$\log(wage_i) = \beta_0 + \beta_1 educ_i + \beta_2 exper_i + \beta_3 exper_i^2 + \beta_4 age_i + \beta_5 kidslt6_i + \beta_6 kidsge6_i + u_i.$$

kidslt6: children under six; *kidsge6*: children aged six or older.

Coefficient	Estimate	Usual SE
Children under six (β_5)	−.06071	.08876

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Coefficient	Estimate	Usual SE
Children under six (β_5)	−.06071	.08876

- Estimate: fitted log-wage association, holding the other regressors fixed.
- SE: estimated sampling spread of this coefficient.

▸ Full model and inference example

Hypothesis testing

Wooldridge 4.2.2 — is the estimate unusual if the coefficient is zero?

Null: $H_0 : \beta_5 = 0$; alternative: $H_1 : \beta_5 \neq 0$.

- Distance from zero in SEs:

$$t = \frac{-.06071}{.08876} \approx -.684.$$

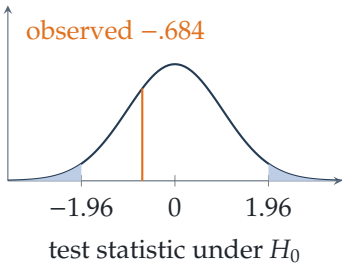
- Null reference: standard normal, approximately.

Hypothesis testing

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- Distance from zero in SEs:
$$t = \frac{-.06071}{.08876} \approx -.684.$$
- Null reference: standard normal, approximately.
- At 5%, reject if $|t| > 1.96$. Here: fail to reject—too little evidence against zero.



▸ Testing, p-values and possible errors

Confidence intervals from the same normal reference

Wooldridge 3.5.1, p. 44; 4.2.2, p. 60 — use the estimate and its standard error

A large-sample 95% confidence interval for β_j is

$$\left[\hat{\beta}_j - 1.96\,\mathrm{se}(\hat{\beta}_j),\ \hat{\beta}_j + 1.96\,\mathrm{se}(\hat{\beta}_j)\right].$$

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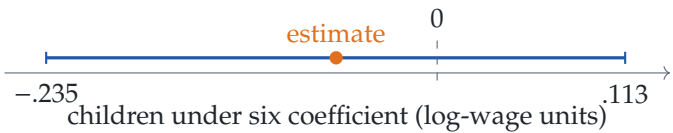
- The normal reference gives approximately 95% coverage across repeated samples.
- The interval changes across samples; the population coefficient is fixed.
- Using the same SE, a value is outside this interval exactly when its two-sided 5% test rejects it.

▸ Calculation and interpretation

Confidence intervals

Wooldridge 3.5.1 and 4.2.2 — show the uncertainty as a range

$$\hat{\beta}_5 \pm 1.96\,\text{se}(\hat{\beta}_5) = -.06071 \pm 1.96(.08876) \approx [-.235,\ .113].$$

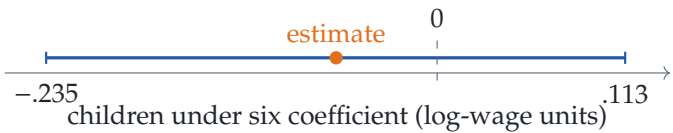


- The interval includes zero and either sign: the estimate is imprecise.

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$$\hat{\beta}_5 \pm 1.96\,\text{se}(\hat{\beta}_5) = -.06071 \pm 1.96(.08876) \approx [-.235,\ .113].$$



- The interval includes zero and either sign: the estimate is imprecise.
- Across repeated samples, this procedure covers the fixed coefficient approximately 95% of the time.

▸ Interval calculation and interpretation

From OLS properties to inference

Wooldridge 4.2.1–4.2.3 — each result has a different job

Maintain OLS.1–OLS.2 and the sampling/moment conditions.

LLN
Consistency
Large misses unlikely.

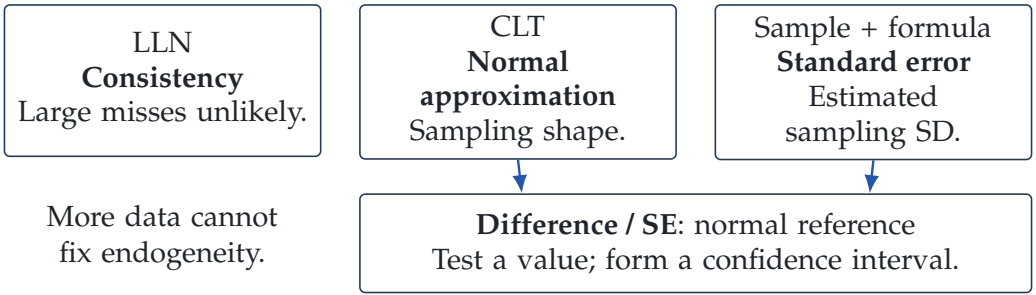
CLT
Normal approximation
Sampling shape.

Sample + formula
Standard error
Estimated sampling SD.

From OLS properties to inference

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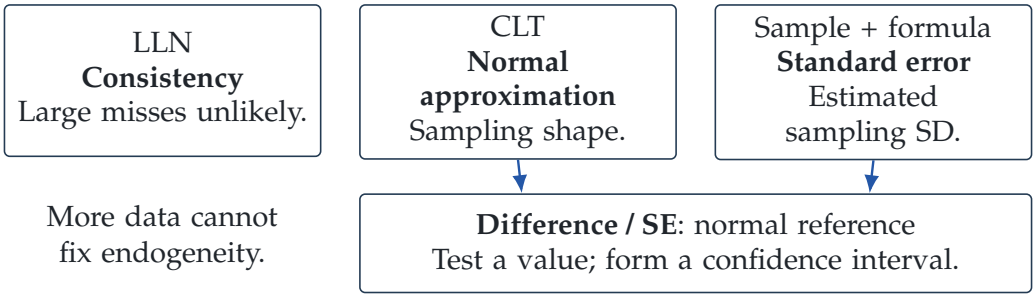


Usual SEs need OLS.3; robust SEs allow its failure.

From OLS properties to inference

Wooldridge 4.2.1–4.2.3 — each result has a different job

Maintain OLS.1–OLS.2 and the sampling/moment conditions.



Usual SEs need OLS.3; robust SEs allow its failure.

MROZ: $t \approx -.684$; no rejection at 5%; zero lies in the 95% CI.

Roadmap

Wooldridge 4.2.3 — current location in blue

OLS recap

4.1 Overview of the Single-Equation Linear Model

4.2 Asymptotic Properties of Ordinary Least Squares

4.2.1 Consistency

4.2.2 Asymptotic Inference Using Ordinary Least Squares

4.2.3 Heteroskedasticity-Robust Inference

Heteroskedasticity-robust standard errors

Wooldridge 4.2.3, pp. 61–62 — retain OLS and estimate its covariance

Without OLS.3, retain the general covariance and estimate $\mathbf{B}$:

$$\text{Avar}(\hat{\boldsymbol{\beta}}) = \frac{\mathbf{A}^{-1}\mathbf{B}\mathbf{A}^{-1}}{N}, \qquad \hat{\mathbf{B}} = \frac{1}{N}\sum_i \hat{u}_i^2 \mathbf{x}_i'\mathbf{x}_i.$$

$$\widehat{\Sigma}_{\text{HC0}} = (\mathbf{X}'\mathbf{X})^{-1}\Big(\sum_i \hat{u}_i^2 \mathbf{x}_i'\mathbf{x}_i\Big)(\mathbf{X}'\mathbf{X})^{-1}. \tag{4.11}$$

Heteroskedasticity-robust standard errors

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- Robust SEs are $\sqrt{[\widehat{\Sigma}]_{jj}}$; OLS coefficients are unchanged.
- Stata: `regress y x1 x2, vce(robust)` uses $\widehat{\Sigma}_{\text{HC1}} = \frac{N}{N-K}\widehat{\Sigma}_{\text{HC0}}$.

▸ MROZ comparison ▸ Covariance derivation

Wrap-up (3a)

Key takeaways

Wooldridge 4.2.1–4.2.3 — estimate, target and uncertainty

- Consistency: Theorem 4.1 says OLS concentrates near its target under OLS.1–OLS.2 and the sampling/moment conditions.

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Wooldridge 4.2.1–4.2.3 — estimate, target and uncertainty

- Consistency: Theorem 4.1 says OLS concentrates near its target under OLS.1–OLS.2 and the sampling/moment conditions.
- Normal approximation: Theorem 4.2 adds OLS.3 for the usual SEs. Normal disturbances are unnecessary; the CLT concerns estimates across samples.
- Inference: combine the estimate with a justified SE for a test or interval. Robust SEs handle changing error spreads, not endogeneity.

▸ The econometric workflow

Next lecture

Wooldridge 4.3–4.4 — when important determinants are missing or mismeasured

- **3b**: more inference, omitted variables and proxies (4.3), then measurement error (4.4).
- Read Sections 4.3–4.4. Detailed inference calculations from today remain in this deck's appendix.
- Problem Set 1 is due Sunday 11 October 2026 (new date).

▸ Appendix contents

Appendix

ECO6102 / Lecture 3

▸ Main lecture: slide 56

Appendix contents

Reference material in textbook order; use the return button after a detour

A	Probability tools and sampling illustrations	App. 1
B	Population restrictions and economic models	App. 16
C	OLS calculations and estimator properties	App. 39
D	Inference calculations and proofs	App. 67
E	Robust covariance and weighting	App. 86
F	Optional LM tests	App. 95

‹ Back to slide 56

Appendix roadmap

APPENDIX A · Wooldridge 3.2–3.5.2 — current location in blue

Chapter 3 tools

The population model and economic interpretation

OLS calculations and estimator properties

Inference: calculations and proofs

Robust covariance and weighting

Optional LM tests

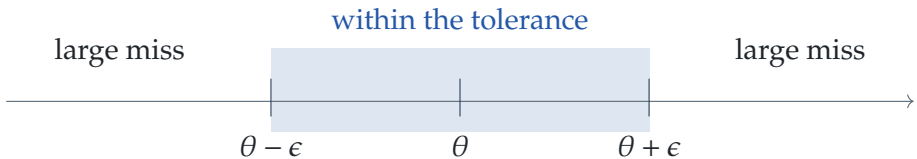
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Consistency

Introductory Wooldridge 6e, C-3a, equation (C.7) — make a large miss unlikely

W_N: estimator from N observations; θ: fixed target; $\epsilon > 0$: fixed tolerance.

$$\Pr\Big(|W_N - \theta| > \epsilon\Big) \longrightarrow 0 \quad \text{for every } \epsilon > 0 \text{ as } N \longrightarrow \infty. \tag{C.7}$$



Equivalently, $W_N \xrightarrow{p} \theta$: the chance of a large miss tends to zero.

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Unbiasedness and consistency

Introductory Wooldridge 6e, C-2d and C-3a — using more observations matters

ILLUSTRATIVE POPULATION

Independent draws $Y_i \sim$ Bernoulli(.5): $E(Y_i) = .5$, $\text{Var}(Y_i) = .25$.

Estimator	Mean	Variance
First observation Y_1	.5	.25
Average $\bar{Y}_N$	.5	$.25/N$

- Both are unbiased. Only the average becomes more precise as N grows.
- Y_1 is always 0 or 1: it remains .5 away from the target, even if we collect more data.

‹ Back to slide 20 › Main lecture: slide 40

The law of large numbers

Introductory Wooldridge 6e, C-3a, equation (C.8) — averages approach population means

Independent, identically distributed observations; finite mean μ.

$$\bar{Y}_N = \frac{1}{N}\sum_{i=1}^{N} Y_i \xrightarrow{p} \mu. \tag{C.8}$$

- An average from a large sample is likely to be close to its population mean.
- An average of 0/1 observations approaches .5 in our example.
- OLS uses such averages. Their limits give consistency under the model's assumptions.

◂ Back to slide 35 ◂ Back to slide 39 ▸ Main lecture: slide 40

The central limit theorem

Introductory Wooldridge 6e, C-3b — a reference shape for sampling uncertainty

Independent, identically distributed observations; mean μ and variance $0 < \sigma^2 < \infty$.

$$\frac{\bar{Y}_N - \mu}{\sigma / \sqrt{N}} \xrightarrow{d} N(0,1).$$

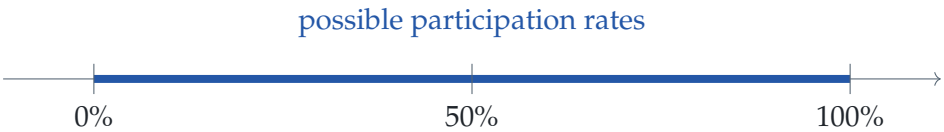
- Center at the target; divide by the sampling SD.
- The resulting sampling distribution approaches the standard normal curve.
- A consistent sample SD can replace the unknown σ.

‹ Back to slide 42

Why normal disturbances can be implausible

Introductory Wooldridge 6e, 5-2, Figure 5.2 — schematic participation-rate support

Participation rate: the percentage of eligible employees in a firm's pension plan.



- Participation cannot be below 0% or above 100%.
- A nondegenerate normal distribution has no such bounds.
- A coefficient estimate can still be approximately normal across samples.

‹ Back to App. 6 › Main lecture: slide 42

Two different distributions

Wooldridge 3.5.1; 4.2.2 — outcomes and estimates vary differently

- **Outcome variation**: among people with the same schooling, wages differ.
- We hold x fixed and vary the person. **Heteroskedasticity**: the conditional disturbance variance changes with x.

▸ A bounded-outcome example

▸ Main lecture: slide 44

- **Sampling variation**: across possible datasets, estimated schooling slopes differ.
- We hold the population and sampling rule fixed and vary the dataset. An SE describes uncertainty about a coefficient estimate.

Convergence in probability

Wooldridge 3.2, Definitions 3.3–3.4, Lemmas 3.2 and 3.4 — plim, o_p, O_p, Slutsky

- $x_N \xrightarrow{p} a$ (**plim** $x_N = a$): for every $\epsilon > 0$, $\Pr(|x_N - a| > \epsilon) \to 0$. Same thing element by element for vectors and matrices.
- $x_N = o_p(1)$: converges in probability to zero. $x_N = O_p(1)$ (**bounded in probability**): for every ϵ there is b_ϵ with $\Pr(|x_N| \geq b_\epsilon) < \epsilon$ for all large N. A plim to any constant implies $O_p(1)$.
- Algebra (Lemma 3.2): $o_p(1) + o_p(1) = o_p(1)$, $O_p(1) + O_p(1) = O_p(1)$, $O_p(1) \cdot O_p(1) = O_p(1)$, $o_p(1) \cdot O_p(1) = o_p(1)$. The last is what makes the $o_p(1)$ in (4.8) harmless.

▸ Main lecture: slide 56

Slutsky's theorem

Wooldridge 3.2, Lemma 3.4, Cor. 3.1 — plims pass through continuous functions

- **Slutsky** (Lemma 3.4): if g is continuous at $\mathbf{c}$ and $\mathbf{x}_N \xrightarrow{p} \mathbf{c}$, then $g(\mathbf{x}_N) \xrightarrow{p} g(\mathbf{c})$: $\operatorname{plim} g(\mathbf{x}_N) = g(\operatorname{plim} \mathbf{x}_N)$. Expectations do not have this property, which is why finite-sample analysis is hard.
- Corollary 3.1: if $\mathbf{Z}_N \xrightarrow{p} \mathbf{A}$ with $\mathbf{A}$ invertible, then $\mathbf{Z}_N^{-1}$ exists with probability approaching one and $\mathbf{Z}_N^{-1} \xrightarrow{p} \mathbf{A}^{-1}$.

▸ Main lecture: slide 56

Convergence in distribution; WLLN and CLT

Wooldridge 3.3–3.4, Definitions 3.6–3.7, Lemmas 3.6–3.7, Theorems 3.1–3.2

- $x_N \xrightarrow{d} x$: the c.d.f. of x_N converges to the (continuous) c.d.f. of x at every point. For vectors: $\mathbf{c}'\mathbf{x}_N \xrightarrow{d} \mathbf{c}'\mathbf{x}$ for every unit vector $\mathbf{c}$ (Cramér–Wold). $\mathbf{x}_N \overset{a}{\sim} N(\boldsymbol{\mu}, \mathbf{V})$ means $\mathbf{x}_N \xrightarrow{d} N(\boldsymbol{\mu}, \mathbf{V})$.
- **Continuous mapping** (Lemma 3.6): $\mathbf{x}_N \xrightarrow{d} \mathbf{x}$ and g continuous imply $g(\mathbf{x}_N) \xrightarrow{d} g(\mathbf{x})$. Corollary 3.2: if $\mathbf{z}_N \xrightarrow{d} N(\mathbf{0}, \mathbf{V})$ with nonsingular $\mathbf{V}$, then $\mathbf{A}'\mathbf{z}_N \xrightarrow{d} N(\mathbf{0}, \mathbf{A}'\mathbf{V}\mathbf{A})$ and $\mathbf{z}_N'\mathbf{V}^{-1}\mathbf{z}_N \xrightarrow{d} \chi^2_K$.
- **Asymptotic equivalence** (Lemma 3.7): if $\mathbf{z}_N \xrightarrow{d} \mathbf{z}$ and $\mathbf{x}_N - \mathbf{z}_N \xrightarrow{p} \mathbf{0}$, then $\mathbf{x}_N \xrightarrow{d} \mathbf{z}$.

▸ Main lecture: slide 56

The two limit theorems

Wooldridge 3.4, Theorems 3.1–3.2 — for random samples

- **WLLN** (Theorem 3.1): $\{\mathbf{w}_i\}$ i.i.d. with finite means: $N^{-1}\sum_i \mathbf{w}_i \xrightarrow{p} E(\mathbf{w}_i)$.
- **CLT** (Theorem 3.2, Lindeberg–Lévy): $\{\mathbf{w}_i\}$ i.i.d. with mean zero and finite second moments: $N^{-1/2}\sum_i \mathbf{w}_i \xrightarrow{d} N(\mathbf{0}, \mathbf{B})$, $\mathbf{B} = E(\mathbf{w}_i\mathbf{w}_i')$. Both survive independent, non-identically distributed draws under weak conditions.

▸ Main lecture: slide 56

Estimators, standard errors, delta method

Wooldridge 3.5.1–3.5.2, Definitions 3.8–3.10, Lemma 3.9, Problem 3.7

- **Consistent**: $\hat{\boldsymbol{\theta}}_N \xrightarrow{p} \boldsymbol{\theta}$ for every value of $\boldsymbol{\theta}$ (3.2). **$\sqrt{N}$-asymptotically normal**: $\sqrt{N}(\hat{\boldsymbol{\theta}}_N - \boldsymbol{\theta}) \xrightarrow{d} N(\mathbf{0}, \mathbf{V})$ (3.3); $\mathbf{V}$ is the asymptotic variance of $\sqrt{N}(\hat{\boldsymbol{\theta}}_N - \boldsymbol{\theta})$ and $\text{Avar}(\hat{\boldsymbol{\theta}}_N) = \mathbf{V}/N$ (3.5). It implies $\hat{\boldsymbol{\theta}}_N - \boldsymbol{\theta} = O_p(N^{-1/2})$: **$\sqrt{N}$-consistent**.
- **Asymptotic standard error** (Definition 3.10): $\text{se}(\hat{\theta}_{Nj}) = (\hat{v}_{Njj}/N)^{1/2}$ with $\hat{\mathbf{V}}_N \xrightarrow{p} \mathbf{V}$ — the number every regression table reports.

▸ Main lecture: slide 56

The delta method, with Problem 3.7

Wooldridge 3.5.2, Lemma 3.9 and (3.10)

- **Delta method** (Lemma 3.9): for $\mathbf{c}(\cdot)$ continuously differentiable with Jacobian $\mathbf{C}(\boldsymbol{\theta})$, $\sqrt{N}[\mathbf{c}(\hat{\boldsymbol{\theta}}_N) - \mathbf{c}(\boldsymbol{\theta})] \overset{a}{\sim} N(\mathbf{0}, \mathbf{CVC}')$ (3.8), so $\widehat{\text{Avar}}[\mathbf{c}(\hat{\boldsymbol{\theta}}_N)] = \hat{\mathbf{C}}_N[\widehat{\text{Avar}}(\hat{\boldsymbol{\theta}}_N)]\hat{\mathbf{C}}_N'$ (3.10), where $\hat{\mathbf{C}}_N = \mathbf{C}(\hat{\boldsymbol{\theta}}_N)$.
- **Problem 3.7**: $\hat{\theta} = 4$, se $= 2$, and $\gamma = \log\theta$. Then $\hat{\gamma} = \log 4 = 1.386$, $\text{se}(\hat{\gamma}) = \text{se}(\hat{\theta})/\hat{\theta} = .5$. Testing $\theta = 1$ gives $t = (4-1)/2 = 1.5$; testing the equivalent $\gamma = 0$ gives $t = 1.386/.5 = 2.77$. Asymptotically equivalent tests can disagree in a given sample.

▸ Main lecture: slide 56

Six words before Chapter 4

Wooldridge 3.5.1, 4.2.1 — six words defined once; each returns where it is used

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▸ Main lecture: slide 9

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▸ Main lecture: slide 9

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Unbiased	right on average at this N: the pile is centred on $\boldsymbol{\beta}$ (4.2.1)
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▸ Main lecture: slide 9

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Estimator	a rule mapping a sample into a coefficient estimate
Analogy principle	population averages → sample averages (**method of moments**; 4.2.1)
Asymptotic	"as $N \to \infty$", used as an approximation at our N (Thm 4.2)
Inference	from one sample to the population, with uncertainty attached (4.2.2–4.2.3)

▸ Main lecture: slide 9

What Lectures 1 and 2 already gave us

Recall — L1: obstacles to causality and random draws; L2: composite slopes

- **L1, obstacles**: omitted variables, measurement error, simultaneity, self-selection — all can make u move with the regressors: endogeneity.

▸ Main lecture: slide 9

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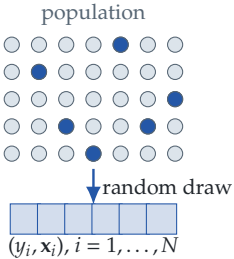
- **L1, obstacles**: omitted variables, measurement error, simultaneity, self-selection — all can make u move with the regressors: endogeneity.
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▸ Main lecture: slide 9

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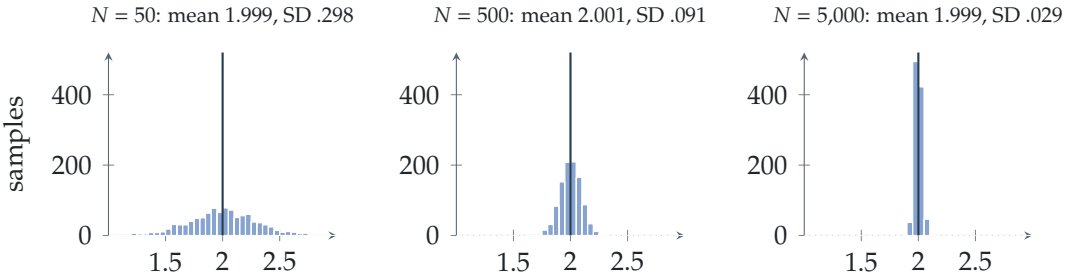
- **L1, obstacles**: omitted variables, measurement error, simultaneity, self-selection — all can make u move with the regressors: endogeneity.
- **L1, random draws**: $(y_i, \mathbf{x}_i)$ are drawn *together*; regressors are random, not fixed.
- **L2, composite slopes**: with ability missing, the observed schooling slope is $\alpha_1 = \beta_1 + \beta_3\delta_1$: target plus ability's link to schooling.



▸ Main lecture: slide 9

A world we built: unbiased and consistent

Wooldridge p. 58 — $y = 3 + 2x + u$: centred on 2 at every N; spreads .298, .091, .029



Known $\beta_1 = 2$: centered at each N, and narrowing. Asymptotic SD: $6/\sqrt{9N} = .283, .089, .028$.

▸ Main lecture: slide 17

Appendix roadmap

APPENDIX B · Wooldridge 4.1 — current location in blue

Chapter 3 tools

The population model and economic interpretation

OLS calculations and estimator properties

Inference: calculations and proofs

Robust covariance and weighting

Optional LM tests

Back to contents

Holding ability fixed versus averaging it out

Wooldridge 4.1 — two conditional means answer different questions

SCHOOLING AND WAGES

$y = \beta_0 + \beta_1 s + \beta_2 e + \gamma q + v$, $E(v \mid s, e, q) = 0$; s schooling, e experience, q ability.

- **Hold ability fixed**:

$$E(y \mid s, e, q) = \beta_0 + \beta_1 s + \beta_2 e + \gamma q.$$

Back to slide 27 Back to contents

Holding ability fixed versus averaging it out

Wooldridge 4.1 — two conditional means answer different questions

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- **Hold ability fixed**:

$$E(y \mid s, e, q) = \beta_0 + \beta_1 s + \beta_2 e + \gamma q.$$

- **Average over ability at each** (s, e):

$$E(y \mid s, e) = \beta_0 + \beta_1 s + \beta_2 e + \gamma E(q \mid s, e).$$

- If ability composition changes with schooling, the observed mean difference includes that change.

◂ Back to slide 27 ◂ Back to contents

Put the omitted variable into the error

Wooldridge 4.1, p. 55 — substitution reveals the covariance that matters

FULL WAGE EQUATION

$y = \beta_0 + \beta_1 educ + \beta_2 exper + \gamma q + v$, with $E(v \mid educ, exper, q) = 0$.

- Without ability in the data, write the same equation as

$$y = \beta_0 + \beta_1 educ + \beta_2 exper + u, \qquad u = \gamma q + v.$$

▸ Main lecture: slide 27

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- Without ability in the data, write the same equation as

$$y = \beta_0 + \beta_1 educ + \beta_2 exper + u, \qquad u = \gamma q + v.$$

- Therefore

$$\mathrm{Cov}(educ, u) = \gamma \, \mathrm{Cov}(educ, q).$$

- The problem arises when ability matters for wages and co-moves with schooling.

▸ Main lecture: slide 27

Measurement error in symbols

Wooldridge 4.4.2 — the same reporting mistake enters the regressor and the error

TRUE AND RECORDED SCHOOLING

$$y = \beta_0 + \beta_1 x^* + v,\ x = x^* + e;\ \mathrm{Cov}(x^*, v) = 0.$$

- **Classical error**: $E(e) = 0$, $\mathrm{Cov}(e, x^*) = \mathrm{Cov}(e, v) = 0$.

‹ Back to slide 29

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- **Classical error**: $E(e) = 0,\ \mathrm{Cov}(e, x^*) = \mathrm{Cov}(e, v) = 0.$
- Substitute recorded schooling:

$$y = \beta_0 + \beta_1 x + u, \quad u = v - \beta_1 e, \quad \mathrm{Cov}(x, u) = -\beta_1 \,\mathrm{Var}(e).$$

▸ Why the simple-regression slope is attenuated

◂ Back to slide 29

When measurement error attenuates the slope

Wooldridge 4.4.2 — a result for classical error in a simple regression

SAME MODEL

$y = \beta_0 + \beta_1 x^* + v,\ x = x^* + e,\ \mathrm{Cov}(x^*, v) = 0.$

- Maintain $E(e) = 0$ and $\mathrm{Cov}(e, x^*) = \mathrm{Cov}(e, v) = 0$, i.i.d. sampling, finite second moments and $\mathrm{Var}(x^*) > 0$.

$$\operatorname{plim} \hat{\beta}_1 = \beta_1 \frac{\mathrm{Var}(x^*)}{\mathrm{Var}(x^*) + \mathrm{Var}(e)}.$$

‹ Back to App. 18 › Main lecture: slide 29

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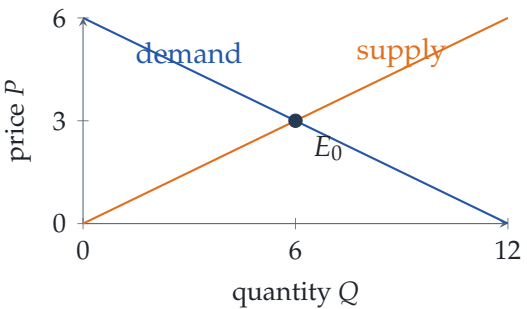
$$\text{plim}\,\hat{\beta}_1 = \beta_1 \frac{\text{Var}(x^*)}{\text{Var}(x^*) + \text{Var}(e)}.$$

- Noise dilutes the signal; more data does not remove the attenuation.
- This is not a universal rule for measurement error in other settings.

‹ Back to App. 18 › Main lecture: slide 29

Demand, supply, and equilibrium

Wooldridge 4.1 — two behavioral relationships jointly determine an observed outcome

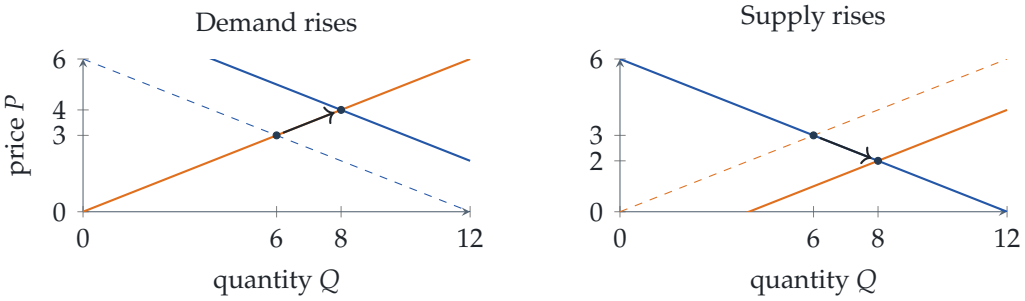


Back to slide 30

- **Demand** (blue): buyers' response to price, holding demand shifters fixed.
- **Supply** (orange): sellers' response, holding supply shifters fixed.
- **Equilibrium**: both choices are compatible at $P = 3$, $Q = 6$.

Which curve moved?

Wooldridge 4.1 — different outside shifts trace different relationships



- Demand shift: $(P, Q) = (3, 6) \rightarrow (4, 8)$; price and quantity rise together.
- Supply shift: $(P, Q) = (3, 6) \rightarrow (2, 8)$; price falls while quantity rises.

▸ Main lecture: slide 30

What a reduced form solves for

Wooldridge 4.1 and 9.2.1 — express equilibrium outcomes in terms of outside determinants

A **reduced form** expresses each jointly determined outcome using only outside determinants.

Behavioral equations $\longrightarrow$ equilibrium $\longrightarrow$ reduced forms.

- In this market, P and Q become functions of the demand and supply shifters, d and s.

‹ Back to slide 33 › Main lecture: slide 30

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Behavioral equations $\longrightarrow$ equilibrium $\longrightarrow$ reduced forms.

- In this market, P and Q become functions of the demand and supply shifters, d and s.
- A regression of Q on jointly determined P is not automatically a reduced form or a demand curve.

‣ Solve the illustrative market

◂ Back to slide 33 ‣ Main lecture: slide 30

Why this course does OLS again

Wooldridge 4.1, p. 53 — fixed regressors to random draws; same formula, new questions

- Some traditional treatments take regressors as fixed. Here $(\mathbf{x}, y)$ is a random draw: both vary across samples. ECO3121 also used random sampling.

▸ Main lecture: slide 9

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▸ Main lecture: slide 9

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- OLS.1–OLS.3 restrict the joint population distribution of $(\mathbf{x}, u)$; conditioning on $\mathbf{X}$ (appendix C) is not $\mathbf{X}$ being fixed by design.
- Statistical questions after identification: whether estimates concentrate near the target (Thm 4.1), how wide and what shape (Thm 4.2), what one sample says about the width (se).

▸ Main lecture: slide 9

Why the zero mean is free

Wooldridge 4.1, p. 54 — with an intercept, $E(u) = 0$ is a normalisation, no more

- Suppose $E(u) = \mu \neq 0$ in (4.1). Set $\beta_0' = \beta_0 + \mu$ and $u' = u - \mu$: the same equation holds with $E(u') = 0$ — same slopes, same y, a relabelled intercept.

◂ Back to slide 24 ◂ Back to App. 27 ▸ Main lecture: slide 21

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- Nothing else changes: $\mathrm{Cov}(x_j, u') = \mathrm{Cov}(x_j, u - \mu) = \mathrm{Cov}(x_j, u)$. Subtracting a constant moves no covariance, so the content of (4.2) is untouched.

◂ Back to slide 24 ◂ Back to App. 27 ▸ Main lecture: slide 21

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- A normalisation, not exogeneity: the intercept absorbs the average of what we left out; nothing about whether it moves with $\mathbf{x}$.
- Without an intercept, mean zero is no longer a free normalization. Zero covariance alone need not make $E(x_j u) = \mathrm{Cov}(x_j, u) + \mu E(x_j)$ zero.

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The conditional mean restriction

Wooldridge 4.1, (4.3)–(4.4) — $E(u \mid \mathbf{x}) = 0$ is sufficient, not necessary, for (4.2)

- A sufficient condition for (4.2) is the zero conditional mean $E(u \mid \mathbf{x}) = 0$ (4.3): then $\mathbf{x}\boldsymbol{\beta}$ is the population regression function $E(y \mid \mathbf{x})$, (4.4).

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‹ Back to slide 24 ‹ Back to slide 25

The conditional mean restriction

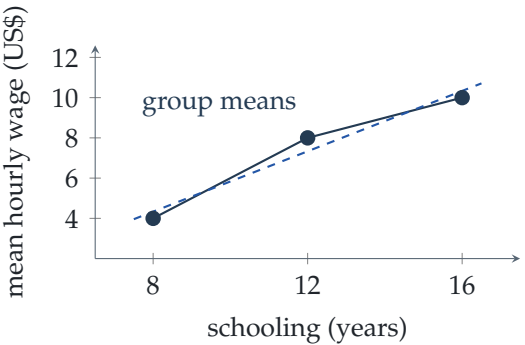
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- It is not necessary: (4.2) alone makes $\mathbf{x}\boldsymbol{\beta}$ the best linear predictor, which can miss the group means (next frame).
- (4.3) says more than (4.2): after including the population line, no additional function of the explanatory variables helps to explain y (p. 54); (4.2) rules out only a linear one.

‹ Back to slide 24 ‹ Back to slide 25

Zero overall is not zero within groups

Wooldridge 4.1 and 2.3 — (4.2) can hold while (4.3) fails: the line misses the means

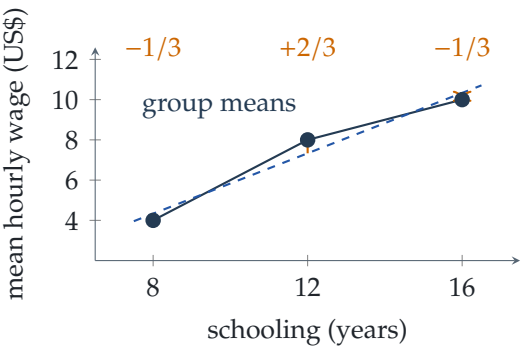


- Three schooling groups, probability 1/3 each; group means $4, 8, 10$.
- Best line: $-5/3 + .75\,x$. Its error v has $E(v) = 0$ and $E(xv) = 0$: (4.2) holds.

▸ Main lecture: slide 24

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- Best line: $-5/3 + .75\,x$. Its error v has $E(v) = 0$ and $E(xv) = 0$: (4.2) holds.
- But $E(v \mid x = 12) = 2/3 \neq 0$: (4.3) fails.

▸ Main lecture: slide 24

The two conditions in words

Wooldridge 4.1, (4.2) and p. 54 — what each condition rules out; linear in parameters

(4.2)

$E(u) = 0$ and $\text{Cov}(x_j, u) = 0$ for every j

- $E(u) = 0$: the intercept absorbs the average of what we left out — a normalisation, free with an intercept.

‣ Appendix B: why the zero mean is free

‣ Main lecture: slide 21

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- $E(u) = 0$: the intercept absorbs the average of what we left out — a normalisation, free with an intercept.
- $\text{Cov}(x_j, u) = 0$: what we left out does not move linearly with what we put in — the content; it is what self-selection breaks.

‣ Appendix B: why the zero mean is free

‣ Main lecture: slide 21

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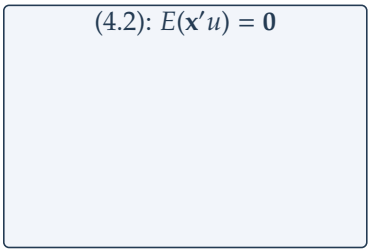
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- $\text{Cov}(x_j, u) = 0$: what we left out does not move linearly with what we put in — the content; it is what self-selection breaks.
- Linear in parameters, not in variables: $\log(income)$, age^2 or $college \cdot age$ as regressors are fine (the book's savings example, p. 54).

‣ Appendix B: why the zero mean is free

‣ Main lecture: slide 21

Three strengths of unrelatedness

Wooldridge pp. 54 and 58 — with $E(u) = 0$, independence is strongest

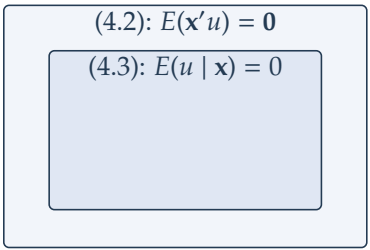


- **(4.2)** (outer): fixes only the covariances; gives the linear projection.

▸ Main lecture: slide 24

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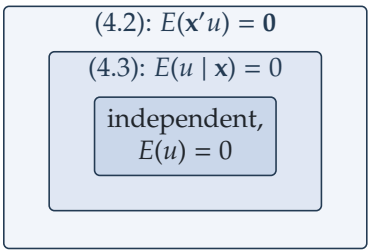


- **(4.3)** (middle): fixes the mean, not the variance; gives (4.4).
- **(4.2)** (outer): fixes only the covariances; gives the linear projection.

▸ Main lecture: slide 24

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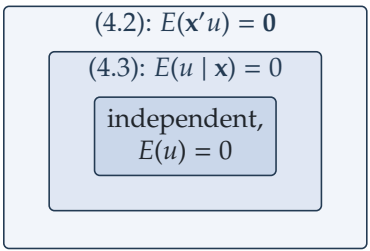


- **Independence** $+E(u) = 0$ (inner): fixes the conditional distribution; implies (4.3) and constant variance.
- **(4.3)** (middle): fixes the mean, not the variance; gives (4.4).
- **(4.2)** (outer): fixes only the covariances; gives the linear projection.

▸ Main lecture: slide 24

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- **Independence** $+E(u) = 0$ (inner): fixes the conditional distribution; implies (4.3) and constant variance.
- **(4.3)** (middle): fixes the mean, not the variance; gives (4.4).
- **(4.2)** (outer): fixes only the covariances; gives the linear projection.

Estimator conclusions retain sampling, rank and moment conditions: (4.3) supports unbiasedness; (4.2) supports consistency.

▸ Main lecture: slide 24

OLS.1 unpacked

Wooldridge 4.2.1 — zero conditional mean gives (4.2), not the other way

- **Element by element**, with $\mathbf{x} = (1, x_1, \ldots, x_p)$: $E(\mathbf{x}'u) = \mathbf{0}$ gives $E(u) = 0$ and $E(x_j u) = \text{Cov}(x_j, u) = 0, j = 1, \ldots, p$.

‹ Back to slide 36

OLS.1 unpacked

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- **Element by element**, with $\mathbf{x} = (1, x_1, \ldots, x_p)$: $E(\mathbf{x}'u) = \mathbf{0}$ gives $E(u) = 0$ and $E(x_j u) = \mathrm{Cov}(x_j, u) = 0$, $j = 1, \ldots, p$.
- **(4.3)** $\Rightarrow$ **(4.2)** by iterated expectations: $E(u) = E[E(u \mid \mathbf{x})] = 0$, $E(x_j u) = E[x_j E(u \mid \mathbf{x})] = 0$; indeed $E[h(\mathbf{x})u] = 0$ for any integrable product $h(\mathbf{x})u$.

‹ Back to slide 36

OLS.1 unpacked

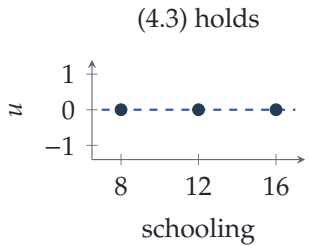
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- **Not the other way**: $x \in \{-1, 0, 1\}$ with probabilities $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$ and $u = x^2 - \frac{1}{2}$: $E(u) = \frac{1}{2} - \frac{1}{2} = 0$, $\mathrm{Cov}(x, u) = E(x^3) = 0$, so (4.2) holds — but $E(u \mid x) = +\frac{1}{2}, -\frac{1}{2}, +\frac{1}{2}$ at $x = -1, 0, 1$: (4.3) fails.

‹ Back to slide 36

What zero covariance looks like

Wooldridge 4.1, (4.2)–(4.3) — u against schooling in three illustrative populations

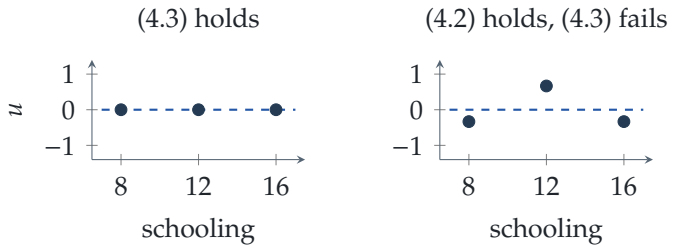


Dashed: the best line through (x, u).

▸ Main lecture: slide 24

What zero covariance looks like

Wooldridge 4.1, (4.2)–(4.3) — u against schooling in three illustrative populations

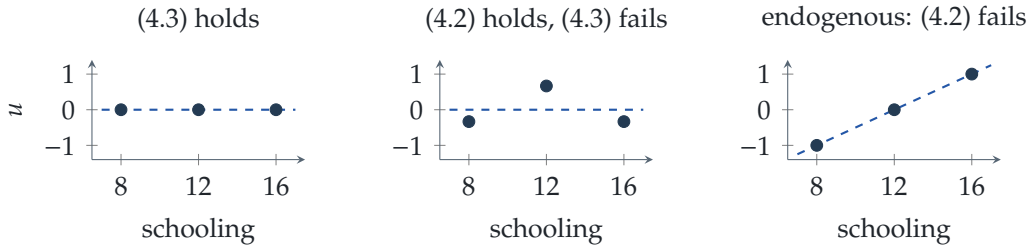


Dashed: the best line through (x, u). Left: every group mean of u is zero. Middle: errors $-1/3,\ 2/3,\ -1/3$: flat line, nonzero group means.

▸ Main lecture: slide 24

What zero covariance looks like

Wooldridge 4.1, (4.2)–(4.3) — u against schooling in three illustrative populations



Dashed: the best line through (x, u). Left: every group mean of u is zero. Middle: errors $-1/3,\ 2/3,\ -1/3$: flat line, nonzero group means. Right: the line tilts; OLS misses β.

▸ Main lecture: slide 24

Two meanings of endogenous

Wooldridge 4.1 and 9.2.1 — distinguish equilibrium determination from regressor–error correlation

- In an economic system, price is **endogenous** because equilibrium determines it jointly with quantity.
- In a regression equation, price is **endogenous** if it is correlated with that equation's disturbance.
- An unobserved demand shock that moves equilibrium price creates this problem in the demand equation.
- If only a supply shifter varies and it is independent of the demand disturbance, joint determination alone does not establish demand-error correlation.

▸ Main lecture: slide 30

Self-selection in the wild

Wooldridge 4.1 — hypothetical applications requested by Jack

- **Job choice and later pay.** If ambition raises both selection into a tech/finance job and later pay, the raw gap can overstate the job effect.

▸ Main lecture: slide 27

Self-selection in the wild

Wooldridge 4.1 — hypothetical applications requested by Jack

- **Job choice and later pay.** If ambition raises both selection into a tech/finance job and later pay, the raw gap can overstate the job effect.
- **AI tools and outcomes.** Better management may raise both adoption and productivity. Learning difficulties may raise AI-tutor use and lower scores. Direction depends on both links.

▸ Main lecture: slide 27

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- **AI tools and outcomes.** Better management may raise both adoption and productivity. Learning difficulties may raise AI-tutor use and lower scores. Direction depends on both links.
- **Training and earnings.** A lay-off may lower earnings and trigger enrolment. A before/after gain may partly reflect recovery from that dip.

▸ Main lecture: slide 27

How an omitted variable becomes part of the error

Wooldridge 4.1, p. 54 — the conditional mean changes when ability is omitted

FULL EQUATION

$y = \beta_0 + \beta_1 educ + \beta_2 exper + \gamma q + v$, with $E(v \mid educ, exper, q) = 0$.

- Absorb unobserved ability: $u = \gamma q + v$. The observed equation is $y = \beta_0 + \beta_1 educ + \beta_2 exper + u$.

▸ Main lecture: slide 27

How an omitted variable becomes part of the error

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- Iterated expectations gives

$$E(u \mid educ, exper) = \gamma E(q \mid educ, exper).$$

▸ Main lecture: slide 27

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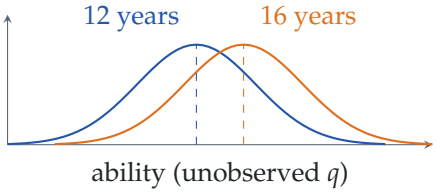
$$E(u \mid educ, exper) = \gamma E(q \mid educ, exper).$$

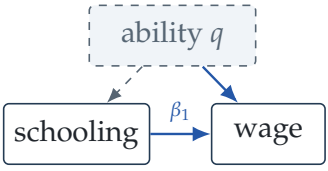
- Centering q makes its overall mean zero; it does not make its mean zero within every schooling/experience group.

▸ Main lecture: slide 27

Self-selection: who picks more schooling?

Wooldridge 4.1, p. 55 — the two schooling groups differ in ability before any wage

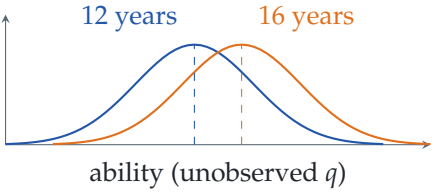


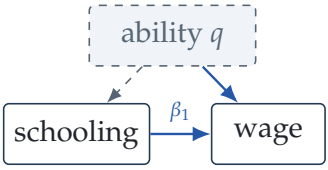


‣ Main lecture: slide 27

Self-selection: who picks more schooling?

Wooldridge 4.1, p. 55 — the two schooling groups differ in ability before any wage



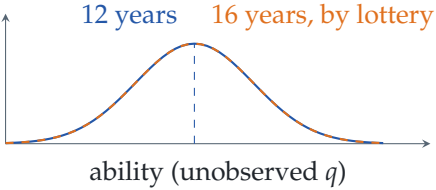


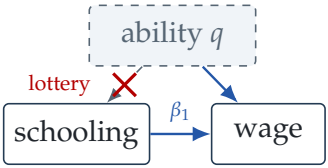
- Wage gap = return to four more years + ability gap between those who chose them (illustrative).

‣ Main lecture: slide 27

Self-selection: who picks more schooling?

Wooldridge 4.1, p. 55 — the two schooling groups differ in ability before any wage





- Wage gap = return to four more years + ability gap between those who chose them (illustrative).
- Random assignment to completed schooling equalizes the population ability distributions and cuts the confounding arrow.

‣ Main lecture: slide 27

Several sources can overlap

Wooldridge 4.1, p. 55 — alcohol and earnings illustrate overlapping concerns

ONE REGRESSION

Suppose we relate earnings to reported alcohol consumption.

- **Omitted factors**: family background or preferences may affect both drinking and earnings.
- **Measurement**: reported consumption can differ from true consumption.
- **Simultaneity**: earnings can affect drinking choices as well as the reverse.

Naming a concern starts the argument; we still need to show how it enters the relevant error.

▸ Main lecture: slide 31

Solve the market and recover its slopes

Wooldridge 4.1 and 9.2.1 — solve the system; recover slopes from shifters

ILLUSTRATIVE EQUATIONS

$Q^d = 12 - 2P + d$, $Q^s = 2P + s$; equilibrium requires $Q^d = Q^s$.

Equating the two gives $12 - 2P + d = 2P + s$, hence

$$P = 3 + \frac{d - s}{4}, \qquad Q = 6 + \frac{d + s}{2}.$$

Demand from a supply shift

$$\frac{\partial Q/\partial s}{\partial P/\partial s} = \frac{1/2}{-1/4} = -2.$$

Supply from a demand shift

$$\frac{\partial Q/\partial d}{\partial P/\partial d} = \frac{1/2}{1/4} = 2.$$

Recovery requires relevant, excluded and exogenous shifters (Lecture 6).

‹ Back to App. 22 › Main lecture: slide 33

Match the method to the policy question

Wooldridge 4.1; Nevo–Whinston (2010) — what can this evidence support?

ONE MARKET, TWO QUESTIONS

A shipping-cost subsidy has been introduced in some markets.

- **Effect of the rollout**: how did prices change because of this policy? We need a credible comparison for what would otherwise have happened.

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Match the method to the policy question

Wooldridge 4.1; Nevo–Whinston (2010) — what can this evidence support?

ONE MARKET, TWO QUESTIONS

A shipping-cost subsidy has been introduced in some markets.

- **Effect of the rollout**: how did prices change because of this policy? We need a credible comparison for what would otherwise have happened.
- **A different policy**: what would a larger subsidy do in another market? An economic model can describe adjustment, but its behavioral and transfer assumptions matter.
- Evidence from a credible rollout can also help identify or assess an economic model.

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Structural and reduced-form research: readings

Optional perspectives on the research approaches introduced beside Wooldridge 4.1

- **Angrist and Pischke (2010): The Credibility Revolution**
  Why credible research design matters.
- **Nevo and Whinston (2010): Taking the Dogma out of Econometrics**
  Structural models, credible inference and counterfactuals.
- **Abadie and Cattaneo (2018): Econometric Methods for Program Evaluation**
  Begin at p. 466, footnote 1: different uses of "reduced form".

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Appendix roadmap

APPENDIX C · Wooldridge 4.2.1–4.2.2 — current location in blue

Chapter 3 tools

The population model and economic interpretation

OLS calculations and estimator properties

Inference: calculations and proofs

Robust covariance and weighting

Optional LM tests

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The vector form with real observations

Wooldridge 4.2, illustration of (4.6) — rows are students; columns are regressors

ATTENDANCE EQUATION

$stndfnl_i = \beta_0 + \beta_1 atndrte_i + u_i.$

- One row is $\mathbf{x}_i = (1, atndrte_i)$, so $\mathbf{x}_i\boldsymbol{\beta} = \beta_0 + \beta_1 atndrte_i$.

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The vector form with real observations

Wooldridge 4.2, illustration of (4.6) — rows are students; columns are regressors

ATTENDANCE EQUATION

$$stndfnl_i = \beta_0 + \beta_1 atndrte_i + u_i.$$

- One row is $\mathbf{x}_i = (1, atndrte_i)$, so $\mathbf{x}_i\boldsymbol{\beta} = \beta_0 + \beta_1 atndrte_i$.
- Three observed rows illustrate the matrix layout:

$$\mathbf{X} = \begin{pmatrix} 1 & 100.0 \\ 1 & 93.8 \\ 1 & 84.4 \end{pmatrix}, \quad \mathbf{Y} = \begin{pmatrix} 2.57 \\ .68 \\ .05 \end{pmatrix}, \quad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}.$$

- Here $K = 2$: the constant and attendance. Estimation uses all observations in the chosen sample.

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The identified population coefficient

Wooldridge 4.2.1, p. 57 — orthogonality plus rank gives one solution

TWO ASSUMPTIONS

OLS.1: $E(\mathbf{x}'u) = \mathbf{0}$; OLS.2: $E(\mathbf{x}'\mathbf{x})$ has full rank K.

- Multiplying $y = \mathbf{x}\boldsymbol{\beta} + u$ by $\mathbf{x}'$ and taking expectations gives

$$E(\mathbf{x}'y) = E(\mathbf{x}'\mathbf{x})\boldsymbol{\beta}.$$

Back to slide 39

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- Multiplying $y = \mathbf{x}\boldsymbol{\beta} + u$ by $\mathbf{x}'$ and taking expectations gives

$$E(\mathbf{x}'y) = E(\mathbf{x}'\mathbf{x})\boldsymbol{\beta}.$$

- OLS.2 allows us to solve:

$$\boldsymbol{\beta} = [E(\mathbf{x}'\mathbf{x})]^{-1}E(\mathbf{x}'y).$$

This is a population result. Its interpretation depends on the equation and orthogonality assumptions we justified.

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The OLS estimator

Wooldridge 4.2.1, p. 57 — the sample analogue of the identified $\boldsymbol{\beta}$

- Replacing the population expectations with sample averages gives the ordinary least squares (OLS) estimator

$$\hat{\boldsymbol{\beta}} = \left[\frac{1}{N}\sum_{i=1}^{N}\mathbf{x}_i'\mathbf{x}_i\right]^{-1}\left[\frac{1}{N}\sum_{i=1}^{N}\mathbf{x}_i'y_i\right] = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}.$$

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Wooldridge 4.2.1, p. 57 — the sample analogue of the identified β

- Replacing the population expectations with sample averages gives the ordinary least squares (OLS) estimator

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- It minimises $\sum_i (y_i - \mathbf{x}_i\mathbf{b})^2$, which is where the name comes from; the moment equations are its first-order conditions.
- $\hat{\boldsymbol{\beta}}$ is random: its value depends on the sample, which is random. How can we tell whether it is a good estimator of $\boldsymbol{\beta}$?

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Why the OLS slope is consistent

Wooldridge 4.2.1, Theorem 4.1 — a sample covariance divided by a sample variance

Consistency: $\operatorname{plim} \hat{\beta}_j = \beta_j$.

SCALAR MODEL

$y_i = \beta_0 + \beta_1 x_i + u_i$; i.i.d. sampling, finite second moments, $E(u) = 0$, $\operatorname{Cov}(x, u) = 0$, and $\operatorname{Var}(x) > 0$.

$$\hat{\beta}_1 - \beta_1 = \frac{N^{-1} \sum_i (x_i - \bar{x})(u_i - \bar{u})}{N^{-1} \sum_i (x_i - \bar{x})^2}$$

Back to slide 40

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$$\hat{\beta}_1 - \beta_1 = \frac{N^{-1} \sum_i (x_i - \bar{x})(u_i - \bar{u})}{N^{-1} \sum_i (x_i - \bar{x})^2} \xrightarrow{p} \frac{\operatorname{Cov}(x, u)}{\operatorname{Var}(x)} = 0.$$

- LLN: sample moments approach population moments.
- Orthogonality: zero numerator. Rank: positive denominator.

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The slope as a covariance ratio

Wooldridge 4.2.1 — a sample identity becomes a population limit

Simple regression with an intercept: $y_i = \beta_0 + \beta_1 x_i + u_i$.

$S_{ab} = N^{-1} \sum_i (a_i - \bar{a})(b_i - \bar{b})$; the fitted sample has $S_{xx} > 0$.

Fitted slope		Target		Sample contribution
$\hat{\beta}_1 = S_{xy}/S_{xx}$	$=$	β_1	$+$	S_{xu}/S_{xx}

Back to slide 5

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Fitted slope		Target		Sample contribution
$\hat{\beta}_1 = S_{xy}/S_{xx}$	=	β_1	+	S_{xu}/S_{xx}

I.i.d. sampling, finite second moments, and $\mathrm{Var}(x) > 0$: LLN gives

$$\operatorname{plim} \hat{\beta}_1 = \beta_1 + \frac{\mathrm{Cov}(x, u)}{\mathrm{Var}(x)}.$$

If $\mathrm{Cov}(x, u) = 0$, the limit is β_1; sample S_{xu} need not be zero.

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When is OLS also unbiased?

Wooldridge 4.2.1, p. 58 and Problem 4.2 — condition on the observed regressor matrix

Unbiasedness: $E(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}$ at a fixed N.

- Add **zero conditional mean** to random sampling: $E(u_i \mid \mathbf{x}_i) = 0$ gives $E(\mathbf{u} \mid \mathbf{X}) = \mathbf{0}$.
- With full sample rank, the regression weights are fixed after conditioning on $\mathbf{X}$:

$$E(\hat{\boldsymbol{\beta}} \mid \mathbf{X}) = \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'E(\mathbf{u} \mid \mathbf{X}) = \boldsymbol{\beta}.$$

Back to slide 40

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$$E(\hat{\boldsymbol{\beta}} \mid \mathbf{X}) = \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'E(\mathbf{u} \mid \mathbf{X}) = \boldsymbol{\beta}.$$

- If the expectation exists, the **law of iterated expectations** gives

$$E(\hat{\boldsymbol{\beta}}) = E\{E(\hat{\boldsymbol{\beta}} \mid \mathbf{X})\} = E(\boldsymbol{\beta}) = \boldsymbol{\beta}.$$

Back to slide 40

OLS first-order conditions

Wooldridge 4.2.1, p. 57 — minimise the squared residuals; the sample moment equations

- One regressor: choose (b_0, b_1) to minimise $\sum_i (y_i - b_0 - b_1 x_i)^2$. The two derivatives set to zero:

$$\sum_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0, \qquad \sum_i x_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0 :$$

  that is $\sum_i \hat{u}_i = 0$ and $\sum_i x_i \hat{u}_i = 0$.

◂ Back to slide 4 ◂ Back to slide 5 ◂ Back to slide 20 ▸ Main lecture: slide 6

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  that is $\sum_i \hat{u}_i = 0$ and $\sum_i x_i \hat{u}_i = 0$.
- Solving: $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$ and $\hat{\beta}_1 = \sum_i (x_i - \bar{x})(y_i - \bar{y}) / \sum_i (x_i - \bar{x})^2$; these also solve the sample moment equations.

◂ Back to slide 4 ◂ Back to slide 5 ◂ Back to slide 20 ▸ Main lecture: slide 6

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- K regressors: minimise $(\mathbf{Y} - \mathbf{Xb})'(\mathbf{Y} - \mathbf{Xb})$; the condition $\mathbf{X}'(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \mathbf{0}$ gives $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$: one orthogonality per column of $\mathbf{X}$.

◂ Back to slide 4 ◂ Back to slide 5 ◂ Back to slide 20 ▸ Main lecture: slide 6

Read the coefficient in its units

Wooldridge 4.2.1 — the ATTEND fitted line, interpreted

$$\widehat{stndfnl} = -.6345 + .00813\, atndrte.$$

- Attendance is measured in percentage points; the final score is standardized.
- Ten more attendance percentage points go with about $.00813 \times 10 = .0813$ standard deviations of score.
- This is a descriptive association. Motivation or prior preparation could affect both attendance and scores.

▸ Population slope formula

▸ Main lecture: slide 8

Identification with one regressor

Wooldridge 4.2.1 — the scalar projection formula

POPULATION MODEL

$y = \beta_0 + \beta_1 x + u$, $E(u) = 0$, $\mathrm{Cov}(x, u) = 0$, $\mathrm{Var}(x) > 0$.

$$\beta_1 = \frac{\mathrm{Cov}(x, y)}{\mathrm{Var}(x)}, \qquad \beta_0 = E(y) - \beta_1 E(x).$$

- The slope scales population co-movement by the variation in x.
- In the ATTEND empirical pool, the projection is $-.6345 + .00813\,atndrte$.
- This pool projection is the known target of the resampling demonstration, not a known causal effect.

‹ Back to App. 46 › Main lecture: slide 39

Think first: we do not have the population

Wooldridge 4.2.1 — from a formula in population moments to a formula in data

IDENTIFIED

$$\boldsymbol{\beta} = [E(\mathbf{x}'\mathbf{x})]^{-1} E(\mathbf{x}'y)$$

- We observe an i.i.d. random sample $\{(\mathbf{x}_i, y_i) : i = 1, \ldots, N\}$, not the population.

▸ Main lecture: slide 38

Method of moments with one regressor

Wooldridge 4.2.1 — replace each population moment by its sample counterpart

POPULATION RESTRICTIONS

$u = y - \beta_0 - \beta_1 x$; $E(u) = 0$ and $E(xu) = 0$.

- Replace expectations by averages and solve

$$\sum_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0, \quad \sum_i x_i (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0.$$

- The solution is the familiar OLS formula:

$$\hat{\beta}_1 = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

▸ Main lecture: slide 39

Why the moment equations identify the coefficients

Wooldridge 4.2.1 — observable population moments and a unique solution

MODEL AND MOMENTS

$y = \mathbf{x}\boldsymbol{\beta} + u$, $E(\mathbf{x}'u) = \mathbf{0}$, $\mathbf{A} = E(\mathbf{x}'\mathbf{x})$ nonsingular.

- Multiply by $\mathbf{x}'$ and take expectations:

$$E(\mathbf{x}'y) = E(\mathbf{x}'\mathbf{x})\boldsymbol{\beta} + E(\mathbf{x}'u) = \mathbf{A}\boldsymbol{\beta}.$$

▸ Main lecture: slide 38

Why the moment equations identify the coefficients

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- Hence $\boldsymbol{\beta} = \mathbf{A}^{-1}E(\mathbf{x}'y)$, a function of population moments of observables.

▸ Main lecture: slide 38

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$$E(\mathbf{x}'y) = E(\mathbf{x}'\mathbf{x})\boldsymbol{\beta} + E(\mathbf{x}'u) = \mathbf{A}\boldsymbol{\beta}.$$

- Hence $\boldsymbol{\beta} = \mathbf{A}^{-1}E(\mathbf{x}'y)$, a function of population moments of observables.
- If $\boldsymbol{\beta}$ and $\boldsymbol{\beta}^*$ both solve the equations, $\mathbf{A}(\boldsymbol{\beta} - \boldsymbol{\beta}^*) = \mathbf{0}$ implies equality. Without rank, uniqueness can fail.

▸ Main lecture: slide 38

Assumption versus sample identity

Wooldridge 4.1–4.2.1 — no hats in the population; hats and subscripts in the sample

	Population (fixed, unknown)	**Sample** (computed, random)
Parameters	$\beta_0, \ldots, \beta_p$	$\hat{\beta}_0, \ldots, \hat{\beta}_p$

Back to slide 36

Assumption versus sample identity

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	Population (fixed, unknown)	**Sample** (computed, random)
Parameters	$\beta_0, \ldots, \beta_p$	$\hat{\beta}_0, \ldots, \hat{\beta}_p$
Line	$E(y \mid \mathbf{x}) = \mathbf{x}\boldsymbol{\beta}$ (under (4.3))	fitted $\hat{y}_i = \mathbf{x}_i\hat{\boldsymbol{\beta}}$

‹ Back to slide 36

Assumption versus sample identity

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	Population (fixed, unknown)	**Sample** (computed, random)
Parameters	$\beta_0, \ldots, \beta_p$	$\hat{\beta}_0, \ldots, \hat{\beta}_p$
Line	$E(y \mid \mathbf{x}) = \mathbf{x}\boldsymbol{\beta}$ (under (4.3))	fitted $\hat{y}_i = \mathbf{x}_i\hat{\boldsymbol{\beta}}$
Deviation	error $u = y - \mathbf{x}\boldsymbol{\beta}$	residual $\hat{u}_i = y_i - \hat{y}_i$

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Assumption versus sample identity

Wooldridge 4.1–4.2.1 — no hats in the population; hats and subscripts in the sample

	Population (fixed, unknown)	**Sample** (computed, random)
Parameters	$\beta_0, \ldots, \beta_p$	$\hat{\beta}_0, \ldots, \hat{\beta}_p$
Line	$E(y \mid \mathbf{x}) = \mathbf{x}\boldsymbol{\beta}$ (under (4.3))	fitted $\hat{y}_i = \mathbf{x}_i\hat{\boldsymbol{\beta}}$
Deviation	error $u = y - \mathbf{x}\boldsymbol{\beta}$	residual $\hat{u}_i = y_i - \hat{y}_i$
Orthogonality	$E(u) = 0$, $E(x_j u) = 0$: assumed	$\sum_i \hat{u}_i = 0$, $\sum_i x_{ij}\hat{u}_i = 0$: always true

The last row: the sample identities hold by construction for any data, so they can never test the population assumption.

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When OLS.2 fails

Wooldridge pp. 57 and 60 — exact linear dependence; why the constant matters

- **The book's examples** (p. 57): every category dummy plus the constant; $\log(age)$ with $\log(age^2) = 2\log(age)$: improper specification (drop one).

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When OLS.2 fails

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- **The book's examples** (p. 57): every category dummy plus the constant; $\log(age)$ with $\log(age^2) = 2\log(age)$: improper specification (drop one).
- **Exact versus high**: only an *exact* relation fails OLS.2; correlation at .95 does not by itself violate OLS.2.

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When OLS.2 fails

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- **The book's examples** (p. 57): every category dummy plus the constant; $\log(age)$ with $\log(age^2) = 2\log(age)$: improper specification (drop one).
- **Exact versus high**: only an *exact* relation fails OLS.2; correlation at .95 does not by itself violate OLS.2.
- **Why the constant matters**: with $\mathbf{x} = (1, \mathbf{z})$, $\mathbf{z}$ the nonconstant regressors,

$$E(\mathbf{x}'\mathbf{x}) = \begin{pmatrix} 1 & E(\mathbf{z}) \\ E(\mathbf{z})' & E(\mathbf{z}'\mathbf{z}) \end{pmatrix}, \qquad E(\mathbf{z}'\mathbf{z}) - E(\mathbf{z})'E(\mathbf{z}) = \mathrm{Var}(\mathbf{z}),$$

  so rank $E(\mathbf{x}'\mathbf{x}) = K \Leftrightarrow \mathrm{Var}(\mathbf{z})$ nonsingular (p. 57); the variance of the full $\mathbf{x}$ is always singular.

‹ Back to slide 37

OLS equals the target plus sampling error

Wooldridge 4.2.1, p. 57 — the identity both theorems of this chapter use

$$\hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + \underbrace{\left[\frac{1}{N}\sum_{i=1}^{N}\mathbf{x}_i'\mathbf{x}_i\right]^{-1}\left[\frac{1}{N}\sum_{i=1}^{N}\mathbf{x}_i'u_i\right]}_{\text{sample average of } \mathbf{x}_i'u_i\text{, rescaled}}$$

- Substitute $y_i = \mathbf{x}_i\boldsymbol{\beta} + u_i$ into the OLS formula and simplify.

▸ Main lecture: slide 39

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- Substitute $y_i = \mathbf{x}_i\boldsymbol{\beta} + u_i$ into the OLS formula and simplify.
- The rightmost bracket is a sample average of the population object whose mean OLS.1 sets to zero. Its probability limit is the key to consistency.

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Why: the plim passes through

Wooldridge 3.2, 3.4, p. 58 — proof of Theorem 4.1: two sample averages, two limits

OLS IDENTITY

$$\hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + \left[\frac{1}{N}\sum_i \mathbf{x}_i'\mathbf{x}_i\right]^{-1}\left[\frac{1}{N}\sum_i \mathbf{x}_i' u_i\right]$$

- **Law of large numbers** (Theorem 3.1): i.i.d. sample averages converge to population means: $\frac{1}{N}\sum_i \mathbf{x}_i'\mathbf{x}_i \xrightarrow{p} E(\mathbf{x}'\mathbf{x})$, $\frac{1}{N}\sum_i \mathbf{x}_i' u_i \xrightarrow{p} E(\mathbf{x}'u) = \mathbf{0}$ (OLS.1).

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Wooldridge 3.2, 3.4, p. 58 — proof of Theorem 4.1: two sample averages, two limits

OLS IDENTITY

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- **Slutsky** (Lemma 3.4): plims pass through continuous functions; inversion is continuous under OLS.2. So the product $\to [E(\mathbf{x}'\mathbf{x})]^{-1}\mathbf{0} = \mathbf{0}$.

▸ Main lecture: slide 40

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- **Slutsky** (Lemma 3.4): plims pass through continuous functions; inversion is continuous under OLS.2. So the product $\to [E(\mathbf{x}'\mathbf{x})]^{-1}\mathbf{0} = \mathbf{0}$.
- For every $\epsilon > 0$, consistency means $\Pr(\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\| > \epsilon) \to 0$. Hence $\operatorname{plim}\hat{\boldsymbol{\beta}} = \boldsymbol{\beta}$: OLS consistently estimates the linear projection of y on $\mathbf{x}$, whatever y is, and the conditional mean when that is linear.

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What Theorem 4.1 does not say

Wooldridge 4.2.1, p. 58 — not causality, not unbiasedness, not accuracy at your N

- It does not say $\boldsymbol{\beta}$ is the parameter you want. If the regressor you care about is endogenous, OLS.1 fails for the structural $\boldsymbol{\beta}$ and OLS converges to something else (3b).

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- It does not say $E(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}$ at any finite N. Under OLS.1–OLS.2 alone the estimator is not necessarily unbiased.

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- It does not say $E(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}$ at any finite N. Under OLS.1–OLS.2 alone the estimator is not necessarily unbiased.
- It does not say how close you are with $N = 20$. The histogram at $N = 20$ had samples with a negative slope.
- It does not assume u independent of $\mathbf{x}$: neither constant conditional variance nor the weaker OLS.3 restriction is needed for consistency.

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Unbiasedness and consistency

Wooldridge 3.5.1; 4.2.1 — centering and concentration answer different questions

OLS with an intercept: maintain the linear model and i.i.d. sampling.

	Unbiasedness	**Consistency**
Meaning	$E(\hat{\beta}_1) = \beta_1$ at a fixed N	$P(\lvert\hat{\beta}_1 - \beta_1\rvert > \epsilon) \to 0$ for every $\epsilon > 0$ as $N \to \infty$
Condition	4 : $E(u \mid x) = 0$	4′ : $E(u) = 0$ and $\mathrm{Cov}(x, u) = 0$
Also require	Full sample rank; $E(\hat{\beta}_1)$ exists.	Population rank; finite second moments.

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Also require	Full sample rank; $E(\hat{\beta}_1)$ exists.	Population rank; finite second moments.

These are sufficient OLS routes. With an intercept, 4′ is OLS.1; $4 \Rightarrow 4'$.

Neither property alone supplies a normal or t reference distribution.

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Unbiasedness under zero conditional mean

Wooldridge p. 58, Problem 4.2 — the introductory finite-sample result, in matrix form

- From the OLS error identity, $\hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{u}$. Under random sampling, $E(u_i \mid \mathbf{X}) = E(u_i \mid \mathbf{x}_i)$, so $E(u \mid \mathbf{x}) = 0$ gives $E(\mathbf{u} \mid \mathbf{X}) = \mathbf{0}$ and

  $$E(\hat{\boldsymbol{\beta}} \mid \mathbf{X}) = \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'E(\mathbf{u} \mid \mathbf{X}) = \boldsymbol{\beta} \quad \text{whenever } \mathbf{X}'\mathbf{X} \text{ is nonsingular.}$$

  If the expectation exists, $E(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}$ by iterated expectations (L2). Adding $\mathrm{Var}(u \mid \mathbf{x}) = \sigma^2$ gives $\mathrm{Var}(\hat{\boldsymbol{\beta}} \mid \mathbf{X}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$.

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Unbiasedness with one regressor

Wooldridge p. 58, Problem 4.2 — the same result with one regressor

MODEL

$y_i = \beta_0 + \beta_1 x_i + u_i, \quad E(u_i \mid \mathbf{X}) = 0$

- **One regressor** (the undergraduate proof): with $w_i = (x_i - \bar{x}) / \sum_j (x_j - \bar{x})^2$,

$$\hat{\beta}_1 = \beta_1 + \sum_i w_i u_i, \qquad E(\hat{\beta}_1 \mid \mathbf{X}) = \beta_1 + \sum_i w_i E(u_i \mid \mathbf{X}) = \beta_1.$$

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Why zero correlation is not enough

Wooldridge Problem 4.3 — and why OLS.3 is written with $E(u^2\mathbf{x}'\mathbf{x})$

- **Why the zero-correlation version does not deliver this** (Problem 4.3): with only $E(\mathbf{x}'u) = \mathbf{0}$, $E(\mathbf{u} \mid \mathbf{X})$ need not be zero, so the conditional mean of $\hat{\boldsymbol{\beta}}$ need not equal $\boldsymbol{\beta}$; and $\text{Var}(u \mid \mathbf{x}) = \sigma^2$ then no longer implies $E(u^2 \mid \mathbf{x}) = \sigma^2$, since $E(u^2 \mid \mathbf{x}) = \sigma^2 + [E(u \mid \mathbf{x})]^2$.

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SLR assumptions versus OLS.1–3

Wooldridge 4.1–4.2.2 — random sampling stays; population moments replace SLR.4–5

- **SLR.1–3**: one regressor is a special case; random sampling stays. OLS.2 requires population rank, not just variation in this sample.

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- **SLR.4**: $E(u \mid x) = 0$ implies OLS.1, $E(\mathbf{x}'u) = 0$. The converse need not hold.

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- **SLR.4**: $E(u \mid x) = 0$ implies OLS.1, $E(\mathbf{x}'u) = 0$. The converse need not hold.
- **SLR.4–5**: zero conditional mean and constant conditional variance imply OLS.3,

$$E(u^2\mathbf{x}'\mathbf{x}) = \sigma^2 E(\mathbf{x}'\mathbf{x}).$$

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Where the root-N equation comes from

Wooldridge 4.2.2, p. 59 — rescale the exact OLS identity

POPULATION MOMENTS

$\mathbf{A} = E(\mathbf{x}'\mathbf{x})$ is nonsingular; $\mathbf{B} = E(u^2\mathbf{x}'\mathbf{x})$, (4.7), is finite.

$$\sqrt{N}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) = \left[\frac{1}{N}\sum_i \mathbf{x}_i'\mathbf{x}_i\right]^{-1}\left[\frac{1}{\sqrt{N}}\sum_i \mathbf{x}_i'u_i\right].$$

- Exact identity: multiply the OLS error identity by $\sqrt{N}$.

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- Exact identity: multiply the OLS error identity by $\sqrt{N}$.
- Under i.i.d. sampling and OLS.1, $\mathrm{Var}(N^{-1/2}\sum_i \mathbf{x}_i'u_i) = \mathbf{B}$: the spread stays finite.

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- The inverse is $\mathbf{A}^{-1} + o_p(1)$; the sum is $O_p(1)$. Their remainder is $o_p(1)$ (Lemma 3.2).

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Proof of Theorem 4.2

Wooldridge 4.2.2, p. 59 — CLT, Slutsky, then the OLS.3 simplification

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- **CLT**: i.i.d. sampling and OLS.1 give $N^{-1/2}\sum_i \mathbf{x}_i' u_i \xrightarrow{d} N(\mathbf{0}, \mathbf{B})$.

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- **Slutsky**: the inverse sample moment converges to $\mathbf{A}^{-1}$, hence

$$\sqrt{N}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) \xrightarrow{d} N(\mathbf{0}, \mathbf{V}), \quad \mathbf{V} = \mathbf{A}^{-1}\mathbf{B}\mathbf{A}^{-1}.$$

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- **OLS.3**: $\mathbf{B} = \sigma^2\mathbf{A}$, so $\mathbf{V} = \sigma^2\mathbf{A}^{-1}$. Otherwise retain the sandwich.

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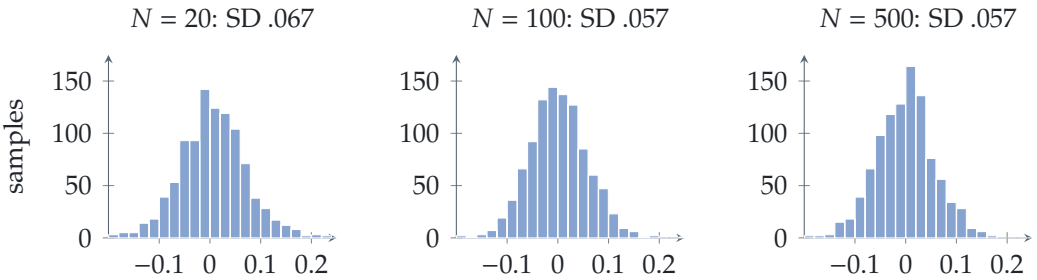
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- **OLS.3**: $\mathbf{B} = \sigma^2\mathbf{A}$, so $\mathbf{V} = \sigma^2\mathbf{A}^{-1}$. Otherwise retain the sandwich.
- $\mathbf{V}$ is the limiting variance of the rescaled estimator; $\mathbf{V}/N$ approximates $\mathrm{Var}(\hat{\boldsymbol{\beta}})$.

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The bell appears

Wooldridge 3.5.1, Def. 3.9 — $\sqrt{N}(\hat{\beta}_1 - \beta_1)$ at $N = 20, 100, 500$: the spread settles



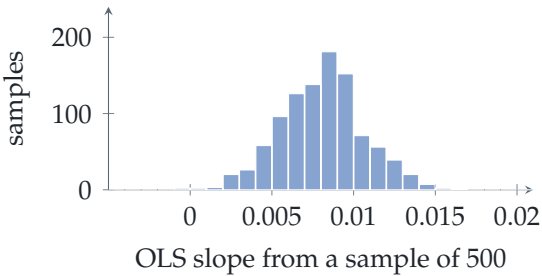
After scaling by $\sqrt{N}$, the SD settles near .0552.

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Standard errors

Wooldridge 4.2.3, (4.11); Definition 3.10 — an SE estimates sampling SD

A standard error estimates sampling spread.



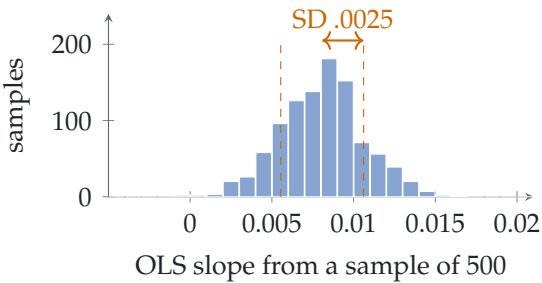
- A **standard error (SE)** estimates how much the coefficient estimate varies across samples.

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Standard errors

Wooldridge 4.2.3, (4.11); Definition 3.10 — an SE estimates sampling SD

A standard error estimates sampling spread.



- A **standard error (SE)** estimates how much the coefficient estimate varies across samples.
- One draw of 500: $\hat{\beta}_1 = .0080$, robust se = .0025.
- The 1,000-draw pile has SD .0025: the SE estimates its width.

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OLS.3 element by element

Wooldridge 4.2.2, p. 59 — with $K = 2$, three scalar statements about u^2

- With $\mathbf{x} = (1, x)$: $E(u^2\mathbf{x}'\mathbf{x}) = \begin{pmatrix} E(u^2) & E(u^2x) \\ E(u^2x) & E(u^2x^2) \end{pmatrix}$ and $\sigma^2 E(\mathbf{x}'\mathbf{x}) = \sigma^2 \begin{pmatrix} 1 & E(x) \\ E(x) & E(x^2) \end{pmatrix}$.

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- $(1,1)$ defines $\sigma^2 = E(u^2)$; $(1,2)$ says $\text{Cov}(u^2, x) = 0$; $(2,2)$ says $\text{Cov}(u^2, x^2) = 0$; in general (j,k) says $\text{Cov}(u^2, x_j x_k) = 0$.

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- **LIE: a sufficient condition**: if $E(u^2 \mid \mathbf{x}) = \sigma^2$, then $E(u^2 x_j x_k) = E[x_j x_k E(u^2 \mid \mathbf{x})] = \sigma^2 E(x_j x_k)$ for every (j,k); it fixes a whole conditional moment, OLS.3 only $K(K+1)/2$ covariances.

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- **Second moment versus variance**: $\mathrm{Var}(u \mid \mathbf{x}) = E(u^2 \mid \mathbf{x}) - [E(u \mid \mathbf{x})]^2$; under (4.3) they coincide, so $E(u^2 \mid \mathbf{x}) = \sigma^2$ reads $\mathrm{Var}(u \mid \mathbf{x}) = \sigma^2$.

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Connect the econometric objects

Wooldridge 4.1–4.2.2 — from the economic question to a qualified conclusion

1. **Target**: define the coefficient and the question it answers.
2. **Model and assumptions**: explain orthogonality, rank, and sampling.
3. **Estimate**: compute $\hat{\boldsymbol{\beta}}$ from the observed data.
4. **Sampling properties**: distinguish unbiasedness, consistency, and normal approximation.
5. **Uncertainty and interpretation**: estimate SEs, conduct inference, and state what the result supports.

3b develops omitted variables, proxies and measurement error.

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Appendix roadmap

APPENDIX D · Wooldridge 4.2.2 — current location in blue

Chapter 3 tools

The population model and economic interpretation

OLS calculations and estimator properties

Inference: calculations and proofs

Robust covariance and weighting

Optional LM tests

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One wage regression for the inference calculations

Wooldridge Example 4.1 — 428 married women in the working sample

$$\log(wage) = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 exper^2 + \beta_4 age + \beta_5 kidslt6 + \beta_6 kidsge6 + u.$$

- *kidslt*6: children under six; *kidsge*6: children aged six or older.

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One wage regression for the inference calculations

Wooldridge Example 4.1 — 428 married women in the working sample

$$\log(wage) = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 exper^2 + \beta_4 age + \beta_5 kidslt6 + \beta_6 kidsge6 + u.$$

- *kidslt*6: children under six; *kidsge*6: children aged six or older.
- Estimate wage associations and their uncertainty.
- Working-sample associations alone do not establish causal effects.

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Read an estimate beside its standard error

Wooldridge 4.2.2; Example 4.1 data — usual standard errors

SAME MODEL

Log wage on education, experience, experience squared, age and both child counts.

Regressor	Estimate	Usual SE
Education	.1078	.0144
Experience	.0398	.0134
Children under six	-.0607	.0888

▸ Main lecture: slide 49

Read an estimate beside its standard error

Wooldridge 4.2.2; Example 4.1 data — usual standard errors

SAME MODEL

Log wage on education, experience, experience squared, age and both child counts.

Regressor	Estimate	Usual SE
Education	.1078	.0144
Experience	.0398	.0134
Children under six	-.0607	.0888

- A negative point estimate can still be imprecise: compare $-.0607$ with its SE, .0888.

▸ Main lecture: slide 49

State the question as a hypothesis

Wooldridge 4.2.2 — a two-sided test of one coefficient

QUESTION

After controlling for the other regressors, is the coefficient on young children zero?

$$H_0 : \beta_5 = 0 \qquad \text{against} \qquad H_1 : \beta_5 \neq 0.$$

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State the question as a hypothesis

Wooldridge 4.2.2 — a two-sided test of one coefficient

QUESTION

After controlling for the other regressors, is the coefficient on young children zero?

$$H_0 : \beta_5 = 0 \qquad \text{against} \qquad H_1 : \beta_5 \neq 0.$$

- Sampling variation can produce a nonzero estimate even if H_0 is true.
- Compare the estimate with the variation we expect *under the null*.

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Measure the distance in standard errors

Wooldridge 4.2.2 — the t statistic has a large-sample normal reference

$$t = \frac{\hat{\beta}_j - \beta_{j,0}}{\text{se}(\hat{\beta}_j)} \overset{a}{\sim} N(0,1) \quad \text{under } H_0.$$

- Children under six:
  $t = -.06071/.08876 \approx -.684.$

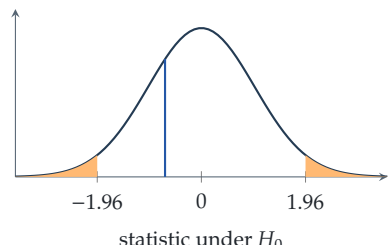
▸ Main lecture: slide 50

Measure the distance in standard errors

Wooldridge 4.2.2 — the t statistic has a large-sample normal reference

$$t = \frac{\hat{\beta}_j - \beta_{j,0}}{\text{se}(\hat{\beta}_j)} \overset{a}{\sim} N(0,1) \quad \text{under } H_0.$$

- Children under six: $t = -.06071/.08876 \approx -.684$.
- At 5%, reject a two-sided null when $|t| > 1.96$.
- Here: fail to reject H_0.



▸ Main lecture: slide 50

A confidence interval shows the uncertainty

Wooldridge 4.2.2 and 3.5.1 — a range of coefficients compatible with this calculation

General rule: $\hat{\beta}_j \pm 1.96 \text{ se}(\hat{\beta}_j)$.

- Children under six:

$$-.06071 \pm 1.96(.08876) \approx [-.235, .113].$$

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A confidence interval shows the uncertainty

Wooldridge 4.2.2 and 3.5.1 — a range of coefficients compatible with this calculation

General rule: $\hat{\beta}_j \pm 1.96 \text{ se}(\hat{\beta}_j)$.

- Children under six:

$$-.06071 \pm 1.96(.08876) \approx [-.235, .113].$$

- Zero and substantial values of either sign are included. Nonrejection does not establish a negligible association.
- Across repeated samples, this procedure covers the fixed coefficient approximately 95% of the time.

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A p-value is a tail probability under the null

Wooldridge 4.2.2 and 3.5.2 — report how unusual the statistic would be

$$p = \Pr\big(|Z| \geq |t_{\text{observed}}|\big), \qquad Z \sim N(0,1).$$

- With $t = -.684$, the two-sided p-value is approximately .494.

‹ Back to slide 48 › Main lecture: slide 50

A p-value is a tail probability under the null

Wooldridge 4.2.2 and 3.5.2 — report how unusual the statistic would be

$$p = \Pr\big(|Z| \geq |t_{\text{observed}}|\big), \qquad Z \sim N(0,1).$$

- With $t = -.684$, the two-sided p-value is approximately .494.
- Reject at level α when $p < \alpha$; $.494 > .05$, so we fail to reject here.
- The p-value is not the probability that H_0 is true.

‹ Back to slide 48 › Main lecture: slide 50

Two possible errors in a testing decision

Wooldridge 3.5.2; 4.2.2 — decisions can be wrong because estimates are random

	H_0 true	H_0 false
Reject H_0	Type I error	Correct rejection
Fail to reject H_0	Correct nonrejection	Type II error

‹ Back to slide 48 › Main lecture: slide 50

Two possible errors in a testing decision

Wooldridge 3.5.2; 4.2.2 — decisions can be wrong because estimates are random

	H_0 true	H_0 false
Reject H_0	Type I error	Correct rejection
Fail to reject H_0	Correct nonrejection	Type II error

- Choose the significance level, such as $\alpha = .05$, before applying the rule.
- A test with asymptotic size .05 falsely rejects with probability approaching .05 under the null.

‹ Back to slide 48 › Main lecture: slide 50

Power is the chance of detecting a false null

Wooldridge 3.5.2; 4.2.2 — precision affects what a test can detect

- **Power**: the probability of rejecting H_0 under a specified alternative.
- A noisy estimate can fail to detect a meaningful coefficient: a Type II error.

▸ Main lecture: slide 50

Power is the chance of detecting a false null

Wooldridge 3.5.2; 4.2.2 — precision affects what a test can detect

- **Power**: the probability of rejecting H_0 under a specified alternative.
- A noisy estimate can fail to detect a meaningful coefficient: a Type II error.
- Under the maintained conditions, larger samples improve detection of a fixed nonzero coefficient; a **consistent test** has power tending to one.

Failure to reject can reflect limited precision. Read the interval as well as the decision.

▸ Main lecture: slide 50

Ask a joint question about age and children

Wooldridge 4.2.2 — test three restrictions together

SAME MROZ MODEL

Education, experience and experience squared remain included.

$$H_0 : \beta_4 = \beta_5 = \beta_6 = 0.$$

- This asks whether age and both child counts jointly add a linear association with wages.

▸ Main lecture: slide 50

Ask a joint question about age and children

Wooldridge 4.2.2 — test three restrictions together

SAME MROZ MODEL

Education, experience and experience squared remain included.

$$H_0 : \beta_4 = \beta_5 = \beta_6 = 0.$$

- This asks whether age and both child counts jointly add a linear association with wages.
- Three separate 5% tests do not generally give one joint test of size 5%. The estimates can move together.

▸ Main lecture: slide 50

Write the joint restrictions as one vector

Wooldridge 4.2.2 — collect the departures from the null

$$\mathbf{d} = \begin{pmatrix} \hat{\beta}_4 \\ \hat{\beta}_5 \\ \hat{\beta}_6 \end{pmatrix}, \qquad \widehat{\Sigma}_d = \text{the corresponding } 3 \times 3 \text{ covariance block.}$$

- Under the joint null, all three population coefficients are zero.

▸ Main lecture: slide 50

Write the joint restrictions as one vector

Wooldridge 4.2.2 — collect the departures from the null

$$\mathbf{d} = \begin{pmatrix} \hat{\beta}_4 \\ \hat{\beta}_5 \\ \hat{\beta}_6 \end{pmatrix}, \qquad \widehat{\Sigma}_d = \text{the corresponding } 3 \times 3 \text{ covariance block.}$$

- Under the joint null, all three population coefficients are zero.
- To measure whether **d** is far from zero, use both its component variances and their covariances.

▸ General restrictions: R beta = r

▸ Main lecture: slide 50

The Wald statistic measures the joint distance

Wooldridge 4.2.2; 3.3 — a covariance-adjusted squared distance

$$W = \mathbf{d}'\widehat{\Sigma}_d^{-1}\mathbf{d} \overset{a}{\sim} \chi^2_Q \quad \text{under } H_0.$$

- Q is the number of independent restrictions. For MROZ, $Q = 3$; the 5% critical value is 7.81.

▸ Main lecture: slide 50

The Wald statistic measures the joint distance

Wooldridge 4.2.2; 3.3 — a covariance-adjusted squared distance

$$W = \mathbf{d}'\widehat{\Sigma}_d^{-1}\mathbf{d} \overset{a}{\sim} \chi^2_Q \quad \text{under } H_0.$$

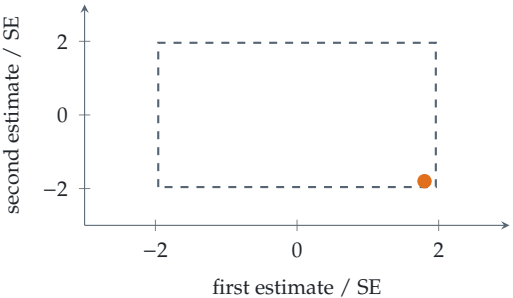
- Q is the number of independent restrictions. For MROZ, $Q = 3$; the 5% critical value is 7.81.
- Reject when $W > 7.81$; the p-value is $\Pr(\chi^2_3 \geq W_{\text{observed}})$.
- With the usual covariance, the usual F statistic is exactly W/Q. The usual covariance is justified by OLS.3.

▸ Why W = t squared, and how F relates

▸ Main lecture: slide 50

Separate tests can miss a joint departure

Wooldridge 4.2.2 — covariance changes which combinations are unusual

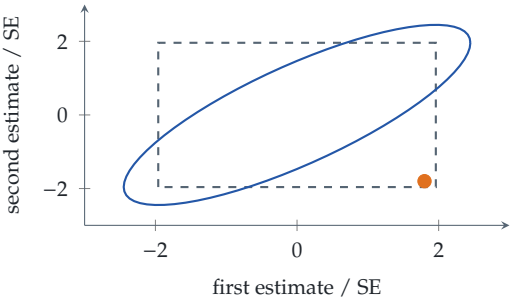


- Illustrative estimates: $(1.8, -1.8)$; both SEs are 1.
- Both individual tests fail to reject: $1.8 < 1.96$ (dashed square).

▸ Main lecture: slide 50

Separate tests can miss a joint departure

Wooldridge 4.2.2 — covariance changes which combinations are unusual



- Illustrative estimates: (1.8, −1.8); both SEs are 1.
- Both individual tests fail to reject: 1.8 < 1.96 (dashed square).
- Estimated correlation .8: opposite movements are unusual. The point lies outside the joint 5% nonrejection ellipse (blue).

▸ Main lecture: slide 50

Residual squares and the error variance

Wooldridge 4.2.2 — use the OLS normal equations

NOTATION

$\boldsymbol{\Delta} = \hat{\boldsymbol{\beta}} - \boldsymbol{\beta}$ and $\hat{\mathbf{u}} = \mathbf{u} - \mathbf{X}\boldsymbol{\Delta}$.

- Expanding the squared residuals gives

$$\frac{\hat{\mathbf{u}}'\hat{\mathbf{u}}}{N} = \frac{\mathbf{u}'\mathbf{u}}{N} - 2\boldsymbol{\Delta}'\frac{\mathbf{X}'\mathbf{u}}{N} + \boldsymbol{\Delta}'\frac{\mathbf{X}'\mathbf{X}}{N}\boldsymbol{\Delta}.$$

‹ Back to slide 46 › Main lecture: slide 49

Residual squares and the error variance

Wooldridge 4.2.2 — use the OLS normal equations

NOTATION

$\boldsymbol{\Delta} = \hat{\boldsymbol{\beta}} - \boldsymbol{\beta}$ and $\hat{\mathbf{u}} = \mathbf{u} - \mathbf{X}\boldsymbol{\Delta}$.

- Expanding the squared residuals gives

$$\frac{\hat{\mathbf{u}}'\hat{\mathbf{u}}}{N} = \frac{\mathbf{u}'\mathbf{u}}{N} - 2\boldsymbol{\Delta}'\frac{\mathbf{X}'\mathbf{u}}{N} + \boldsymbol{\Delta}'\frac{\mathbf{X}'\mathbf{X}}{N}\boldsymbol{\Delta}.$$

- The OLS normal equations imply $\mathbf{X}'\mathbf{u} = \mathbf{X}'\mathbf{X}\boldsymbol{\Delta}$, so

$$\frac{SSR}{N} = \frac{1}{N}\sum_i u_i^2 - \boldsymbol{\Delta}'\Big(\frac{\mathbf{X}'\mathbf{X}}{N}\Big)\boldsymbol{\Delta}.$$

‹ Back to slide 46 › Main lecture: slide 49

Consistency of the residual variance

Wooldridge 4.2.2 — consistency is a large-sample statement

$$\frac{SSR}{N} = \frac{1}{N}\sum_i u_i^2 - \boldsymbol{\Delta}'\Big(\frac{\mathbf{X}'\mathbf{X}}{N}\Big)\boldsymbol{\Delta}.$$

- I.i.d. sampling and finite second moments give $N^{-1}\sum_i u_i^2 \xrightarrow{p} E(u^2)$.
- Consistency: $\boldsymbol{\Delta} \xrightarrow{p} 0$. Since $\mathbf{X}'\mathbf{X}/N \xrightarrow{p} A$, the quadratic term tends to zero.

▸ Main lecture: slide 49

Consistency of the residual variance

Wooldridge 4.2.2 — consistency is a large-sample statement

$$\frac{SSR}{N} = \frac{1}{N}\sum_i u_i^2 - \boldsymbol{\Delta}'\Big(\frac{\mathbf{X}'\mathbf{X}}{N}\Big)\boldsymbol{\Delta}.$$

- I.i.d. sampling and finite second moments give $N^{-1}\sum_i u_i^2 \xrightarrow{p} E(u^2)$.
- Consistency: $\boldsymbol{\Delta} \xrightarrow{p} 0$. Since $\mathbf{X}'\mathbf{X}/N \xrightarrow{p} A$, the quadratic term tends to zero.
- Fixed K: $N/(N-K) \to 1$, hence $s^2 \xrightarrow{p} E(u^2)$.

This consistency result needs no OLS.3; it does not establish unbiasedness.

▸ Main lecture: slide 49

Studentizing the estimator

Wooldridge 3.5.1–3.5.2; 4.2.2 — estimate the scale consistently

- Suppose $\sqrt{N}(\hat{\beta}_j - \beta_j) \xrightarrow{d} N(0, V_{jj})$, with $V_{jj} > 0$, and $\hat{V}_{jj} \xrightarrow{p} V_{jj}$.

‹ Back to slide 47 › Main lecture: slide 50

Studentizing the estimator

Wooldridge 3.5.1–3.5.2; 4.2.2 — estimate the scale consistently

- Suppose $\sqrt{N}(\hat{\beta}_j - \beta_j) \xrightarrow{d} N(0, V_{jj})$, with $V_{jj} > 0$, and $\hat{V}_{jj} \xrightarrow{p} V_{jj}$.
- Slutsky's theorem gives

$$\frac{\hat{\beta}_j - \beta_j}{\sqrt{\hat{V}_{jj}/N}} \xrightarrow{d} N(0, 1).$$

- Under H_0, substitute $\beta_{j,0}$ in the numerator. Invert this test to obtain the confidence interval.

The lecture uses a large-sample normal reference.

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Construct a restriction matrix

Wooldridge 4.2.3, (4.12) — R selects or combines coefficients

$$H_0 : R\boldsymbol{\beta} = r. \tag{4.12}$$

R is $Q \times K$ with rank Q; r is $Q \times 1$.

- For the MROZ joint null:

$$R = \begin{pmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}, \qquad r = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

◂ Back to App. 76 ▸ Main lecture: slide 50

Construct a restriction matrix

Wooldridge 4.2.3, (4.12) — R selects or combines coefficients

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- For the MROZ joint null:

$$R = \begin{pmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}, \qquad r = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

- For $\beta_j = \beta_k$, use one row with 1 at j, -1 at k, and zeros elsewhere.

◂ Back to App. 76 ▸ Main lecture: slide 50

Wald, t squared and the usual F statistic

Wooldridge 3.3; adapted (4.13) — covariance normalizes the restrictions

$$W = (R\hat{\boldsymbol{\beta}} - r)'[R\widehat{\Sigma}R']^{-1}(R\hat{\boldsymbol{\beta}} - r) \overset{a}{\sim} \chi^2_Q.$$

- A standardized Q-dimensional normal vector has a chi-square squared length. A consistent covariance estimate gives the same limit.

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Wald, t squared and the usual F statistic

Wooldridge 3.3; adapted (4.13) — covariance normalizes the restrictions

$$W = (R\hat{\boldsymbol{\beta}} - r)'[R\widehat{\Sigma}R']^{-1}(R\hat{\boldsymbol{\beta}} - r) \overset{a}{\sim} \chi^2_Q.$$

- A standardized Q-dimensional normal vector has a chi-square squared length. A consistent covariance estimate gives the same limit.
- With one restriction, $W = t^2$ using the same covariance estimate.
- With $\widehat{\Sigma} = s^2(\mathbf{X}'\mathbf{X})^{-1}$, $F = W/Q$ exactly. Under the classical normal model $F \sim F(Q, N-K)$; without it the justification here is asymptotic.

◂ Back to App. 77 ▸ Main lecture: slide 50

The full MROZ coefficient table

Wooldridge Example 4.1 — all 428 observations enter both calculations

Regressor	Coefficient	Usual SE	HC1 SE
Intercept	-.42091	.31690	.31833
Education	.10783	.01440	.01362
Experience	.03982	.01339	.01526
Experience squared	-.000781	.000402	.000410
Age	-.001465	.005292	.005935
Children under six	-.06071	.08876	.10610
Older children	-.01459	.02790	.02935

$N = 428$, $K = 7$, $R^2 = .15824$; one OLS fit.

▸ Main lecture: slide 49

Reproduce the inference calculation in Python

Wooldridge Example 4.1 — the companion script exposes each calculation

1. Select working women and the common complete sample; cast experience to floating point before squaring it.
2. Fit OLS once; calculate usual and HC1 covariance matrices from the residuals.
3. Use normal references for individual tests and intervals; use χ^2_3 for the Wald tests.

‹ Back to App. 91 › Main lecture: slide 54

Reproduce the inference calculation in Python

Wooldridge Example 4.1 — the companion script exposes each calculation

1. Select working women and the common complete sample; cast experience to floating point before squaring it.
2. Fit OLS once; calculate usual and HC1 covariance matrices from the residuals.
3. Use normal references for individual tests and intervals; use χ^2_3 for the Wald tests.

Companion file: `L3a_inference_mroz.py`. Output includes every coefficient, SE, interval, p-value and the joint tests.

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Appendix roadmap

APPENDIX E · Wooldridge 4.2.3 — current location in blue

Chapter 3 tools

The population model and economic interpretation

OLS calculations and estimator properties

Inference: calculations and proofs

Robust covariance and weighting

Optional LM tests

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Heteroskedasticity-robust inference

Wooldridge 4.2.3 — keep the OLS estimate and adjust its uncertainty

- If the disturbance spread changes across regressors, the usual SE can be misleading.

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Heteroskedasticity-robust inference

Wooldridge 4.2.3 — keep the OLS estimate and adjust its uncertainty

- If the disturbance spread changes across regressors, the usual SE can be misleading.
- Robust SEs allow that spread to vary. Keep the OLS coefficient; use the revised SE in the same tests and intervals.

Estimate	Usual SE	Robust SE (HC1)
−.06071	.08876	.10610

Robust SEs address heteroskedasticity; they do not remove endogeneity.

▸ Robust covariance example

◂ Back to slide 54 ◂ Back to contents

The same estimates with robust standard errors

Wooldridge 4.2.3; Example 4.1 data — HC1 rescales the sandwich

$$\widehat{\Sigma}_{\text{HC1}} = \frac{N}{N-K}\widehat{\Sigma}_{\text{HC0}}.$$

Regressor	Estimate	Usual SE	HC1 SE
Education	.1078	.0144	.0136
Experience	.0398	.0134	.0153
Children under six	-.0607	.0888	.1061

‹ Back to App. 86 › Main lecture: slide 54

The same estimates with robust standard errors

Wooldridge 4.2.3; Example 4.1 data — HC1 rescales the sandwich

$$\widehat{\Sigma}_{\text{HC1}} = \frac{N}{N-K}\widehat{\Sigma}_{\text{HC0}}.$$

Regressor	Estimate	Usual SE	HC1 SE
Education	.1078	.0144	.0136
Experience	.0398	.0134	.0153
Children under six	-.0607	.0888	.1061

- For children under six, robust $t = -.572$ and $p = .567$: still fail to reject.

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Make the joint test robust in the same way

Wooldridge 4.2.3, adapted (4.13) with HC1 — keep the hypothesis; replace its covariance

$$W_{\text{robust}} = \mathbf{d}'[\widehat{\Sigma}_{d,\text{HC1}}]^{-1}\mathbf{d} \overset{a}{\sim} \chi^2_Q \quad \text{under } H_0.$$

- The restriction vector and the OLS estimates stay the same.

‣ Main lecture: slide 54

Make the joint test robust in the same way

Wooldridge 4.2.3, adapted (4.13) with HC1 — keep the hypothesis; replace its covariance

$$W_{\text{robust}} = \mathbf{d}'[\widehat{\Sigma}_{d,\text{HC1}}]^{-1}\mathbf{d} \overset{a}{\sim} \chi^2_Q \quad \text{under } H_0.$$

- The restriction vector and the OLS estimates stay the same.
- Use the HC1 covariance block. The joint chi-square reference remains valid under the maintained sampling and moment conditions.

▸ Main lecture: slide 54

Robust inference and weighted least squares have different jobs

Wooldridge 4.2.3 — variance estimation versus changing the estimator

- **Robust SEs**: retain the OLS fit and allow heteroskedasticity in its uncertainty calculation.

‣ Main lecture: slide 54

Robust inference and weighted least squares have different jobs

Wooldridge 4.2.3 — variance estimation versus changing the estimator

- **Robust SEs**: retain the OLS fit and allow heteroskedasticity in its uncertainty calculation.
- **Weighted least squares**: change the fit by giving observations different weights. Efficiency gains require a suitable conditional-mean and variance model.
- HC1 is a finite-sample adjustment, not a guarantee of accurate small-sample inference.

▸ When weighting helps or changes the target ▸ Small-sample qualifications

▸ Main lecture: slide 54

The joint MROZ decision

Wooldridge 4.2.3; Example 4.1 data — the null concerns age and both child counts

JOINT NULL

$H_0 : \beta_4 = \beta_5 = \beta_6 = 0$, with $Q = 3$.

Covariance	Wald W	p from χ^2_3
Usual	.711	.871
HC1	.493	.920

- The usual $F = W/3 = .237$; using $F(3, 421)$ also gives $p \approx .871$.

▸ Main lecture: slide 54

The joint MROZ decision

Wooldridge 4.2.3; Example 4.1 data — the null concerns age and both child counts

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$H_0 : \beta_4 = \beta_5 = \beta_6 = 0$, with $Q = 3$.

Covariance	Wald W	p from χ^2_3
Usual	.711	.871
HC1	.493	.920

- The usual $F = W/3 = .237$; using $F(3, 421)$ also gives $p \approx .871$.
- Both Wald statistics are below 7.81: fail to reject. The data do not establish that all three coefficients are zero.

▸ Main lecture: slide 54

From a dataset to an inference statement

Wooldridge 4.2.2–4.2.3 — one fit, explicit assumptions, a qualified conclusion

1. Specify the sample, model and hypothesis; fit OLS once.
2. Choose a justified covariance estimate and compute the statistic or interval.
3. Report the estimate, uncertainty and what the evidence supports.

▸ Main lecture: slide 54

From a dataset to an inference statement

Wooldridge 4.2.2–4.2.3 — one fit, explicit assumptions, a qualified conclusion

1. Specify the sample, model and hypothesis; fit OLS once.
2. Choose a justified covariance estimate and compute the statistic or interval.
3. Report the estimate, uncertainty and what the evidence supports.

More precise estimates do not resolve an omitted-variable problem. Section 4.3 asks what happens when an important determinant is unobserved.

‣ Python walkthrough and its reference conventions

‣ Main lecture: slide 54

Why the usual covariance is a special case

Wooldridge 4.2.3 — simplify population moments, not individual residual squares

$$A = E(\mathbf{x}'\mathbf{x}), \quad B = E(u^2\mathbf{x}'\mathbf{x}), \quad V = A^{-1}BA^{-1}, \quad \Sigma = V/N.$$

- Under OLS.3, $B = \sigma^2 A$, hence $V = \sigma^2 A^{-1}$.

‹ Back to slide 54

Why the usual covariance is a special case

Wooldridge 4.2.3 — simplify population moments, not individual residual squares

$$A = E(\mathbf{x}'\mathbf{x}), \quad B = E(u^2\mathbf{x}'\mathbf{x}), \quad V = A^{-1}BA^{-1}, \quad \Sigma = V/N.$$

- Under OLS.3, $B = \sigma^2 A$, hence $V = \sigma^2 A^{-1}$.
- In data use $\hat{A} = \mathbf{X}'\mathbf{X}/N$ and $\hat{B} = N^{-1}\sum_i \hat{u}_i^2 \mathbf{x}_i'\mathbf{x}_i$ for the sandwich.
- The simplification concerns population moments. Homoskedasticity does not say that every fitted residual square equals s^2.

‹ Back to slide 54

When weighted least squares helps

Wooldridge 4.2.3 — weighting needs its own justification

- With a correct conditional mean and variance model, inverse-variance weighting can improve efficiency by downweighting noisier observations.

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When weighted least squares helps

Wooldridge 4.2.3 — weighting needs its own justification

- With a correct conditional mean and variance model, inverse-variance weighting can improve efficiency by downweighting noisier observations.
- WLS changes the coefficient estimate; robust standard errors leave OLS coefficients unchanged.
- If only $E(\mathbf{x}'u) = 0$ holds, weighting may fail $E[w(\mathbf{x})\mathbf{x}'u] = 0$ and change the projection target.

Correct weighting is an additional modeling task. Chapter 12 develops these methods further.

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What HC1 does and does not promise

Wooldridge 4.2.3 — the correction changes covariance by N divided by N minus K

$$\widehat{\Sigma}_{\mathrm{HC1}} = \frac{N}{N-K}\widehat{\Sigma}_{\mathrm{HC0}}, \qquad \frac{428}{421} \approx 1.0166.$$

- This degrees-of-freedom adjustment can improve small-sample approximation, but it does not make robust tests exact.

‹ Back to App. 89 › Main lecture: slide 54

What HC1 does and does not promise

Wooldridge 4.2.3 — the correction changes covariance by N divided by N minus K

$$\widehat{\Sigma}_{\mathrm{HC1}} = \frac{N}{N-K}\widehat{\Sigma}_{\mathrm{HC0}}, \qquad \frac{428}{421} \approx 1.0166.$$

- This degrees-of-freedom adjustment can improve small-sample approximation, but it does not make robust tests exact.
- The textbook discusses finite-sample adjustments to robust covariance estimates, including HC1.
- Heteroskedasticity robustness leaves exogeneity, sampling and moment requirements in place.

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Appendix roadmap

APPENDIX F · Wooldridge 4.2.4 — current location in blue

Chapter 3 tools

The population model and economic interpretation

OLS calculations and estimator properties

Inference: calculations and proofs

Robust covariance and weighting

Optional LM tests

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The LM idea: look for what the restricted model misses

Wooldridge 4.2.4 — start by imposing the joint null

1. Fit the wage equation without age and the two child counts: impose H_0.
2. Keep the restricted residuals, $\tilde{u}_i$.

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The LM idea: look for what the restricted model misses

Wooldridge 4.2.4 — start by imposing the joint null

1. Fit the wage equation without age and the two child counts: impose H_0.
2. Keep the restricted residuals, $\tilde{u}_i$.
3. Ask whether the excluded regressors explain these residuals after accounting for the included regressors.

The LM test uses a restricted fit. The Wald test measures the unrestricted estimate's departure from the null.

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The ordinary LM statistic

Wooldridge 4.2.4, (4.14)–(4.15) — the auxiliary regression must include all regressors

- Regress the restricted residuals $\tilde{u}$ on *both* the retained regressors $\mathbf{X}_1$ and the excluded regressors $\mathbf{X}_2$.
- With its R_u^2, the statistic is

$$LM = NR_u^2 \overset{a}{\sim} \chi_Q^2 \quad \text{under } H_0 \text{ and OLS.3.}$$

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The ordinary LM statistic

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- MROZ: $R_u^2 = .001686$, $LM = .722$, $p = .868$: fail to reject the three restrictions.

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The heteroskedasticity-robust LM calculation

Wooldridge 4.2.4, (4.17) — partial out the retained regressors first

1. Regress each excluded regressor on $\mathbf{X}_1$; collect the residuals $\hat{\mathbf{r}}_i$.
2. Regress the constant 1 on the products $\tilde{u}_i\hat{\mathbf{r}}_i$, *without an intercept*; call its sum of squared residuals SSR_0.

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The heteroskedasticity-robust LM calculation

Wooldridge 4.2.4, (4.17) — partial out the retained regressors first

1. Regress each excluded regressor on $\mathbf{X}_1$; collect the residuals $\hat{\mathbf{r}}_i$.
2. Regress the constant 1 on the products $\tilde{u}_i \hat{\mathbf{r}}_i$, *without an intercept*; call its sum of squared residuals SSR_0.
3. Use $LM_{\text{robust}} = N - SSR_0 \overset{a}{\sim} \chi^2_Q$ under H_0 and the maintained moment conditions.

MROZ: $LM_{\text{robust}} = .512$, $p = .916$. The robust Wald and LM tests agree on nonrejection here.

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